

Monopsony Power and the Transmission of Monetary Policy*

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Abstract

This paper studies how labor market power affects the transmission of monetary policy. Using administrative U.S. Census data, we show that firms with high monopsony power—defined as those accounting for over 10 percent of the local wage bill—respond less to monetary policy in terms of their wage bill and employment. We then develop a New Keynesian model with heterogeneous firms and oligopsonistic competition to interpret these findings. Wage stickiness combined with firms’ labor market power is key to generating the heterogeneous responses that we document. Our model highlights two channels through which oligopsony shapes the aggregate effects of monetary policy: partial passthrough and misallocation. Calibrated to U.S. labor markets, the model implies that the decline in labor market power since the 1980s has increased the output response to monetary policy by about 10 percent and accounts for about 15 percent of the estimated flattening of the Phillips curve.

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1 Introduction

Over the past four decades, local labor markets in the U.S. have become less concentrated.¹ The average Herfindahl-Hirschman index of local labor market wage bills has declined from 0.35 in 1980 to 0.24 in 2019.² Despite this decline, local labor markets remain highly concentrated: less than 6 percent of firms account for 40 percent of employment. This high degree of concentration can lead to a partial and heterogeneous pass-through of demand shocks to wages, a key element in the transmission of monetary policy to real outcomes.

In this paper, we study how labor market power shapes the transmission of monetary policy. We address this question both empirically and theoretically. On the empirical side, we use administrative U.S. Census data to document the heterogeneous response of firms to a monetary policy shock. We classify firms into low- and high-monopsony-power groups, according to their wage-bill share in the local labor market. Our analysis shows that low-monopsony-power firms are significantly more responsive to monetary policy shocks in terms of their wage bill and their employment.

On the theoretical side, we construct a heterogeneous agents New Keynesian model with oligopsonistic competition. The economy consists of a continuum of local labor markets, and within each labor market, a finite number of heterogeneous firms employ workers. These firms compete à la Bertrand and can adjust their nominal wages infrequently. Our model implies two channels through which oligopsonistic competition shapes the aggregate effects of monetary policy: a *partial passthrough channel* and a *misallocation channel*. We develop a numerical method to solve the model, and calibrate the model to match key features of the U.S. labor market. In the calibrated model, like in the data, low-monopsony-power firms are more responsive to monetary policy. Quantitatively, we find that oligopsonistic competition dampens the effect of monetary policy on output by 12 percent, relative to an economy where firms strive to keep their markdowns constant. Furthermore, the decline in labor market concentration since 1980 has amplified the transmission of monetary policy to aggregate output by about 10 percent.

The dataset we use to study the heterogeneous response of firms to monetary policy is the Census’ Longitudinal Employer-Household Dynamics (LEHD) merged with the Longitudinal Business Database (LBD). The LEHD contains quarterly information on wage bill and employment of all establishments with employees in the U.S. between 1993 and 2019, including the county in which they are located, their age, industry, and the firm to which the establishment belongs. We follow [Yeh, Macaluso, and Hershbein \(2022\)](#) in defining a local labor market as all workers employed in firms in the same subsector (3-digit NAICS) in the same county. Then, we classify a firm-county-subsector triplet as a high-monopsony-power firm if it accounts for more than 10 percent of the total wage bill in the local labor market. Otherwise, the firm is classified as a low-monopsony-power firm.

We use local projection methods and high-frequency monetary policy *shocks* to identify the dynamic effects of monetary policy on firms’ wage bill. Our main empirical finding is that low-monopsony-power firms are more responsive to a change in monetary policy. In response to an unexpected decline

¹See [Rossi-Hansberg et al. \(2021\)](#) and [Berger et al. \(2022\)](#).

²Authors’ calculation using the U.S. Census Longitudinal Business Database. We follow [Yeh et al. \(2022\)](#) in defining a local labor market as a subsector (3-digit NAICS) in a county.

in the federal funds rate of 25 basis points, the wage bill of low-monopsony-power firms responds by 0.6 log points more than that of high-monopsony-power firms in the second year following the shock. To put this number in perspective, while both groups of firms expand wage bill following an expansionary monetary policy shock, low-monopsony-power firms' response is about 40 percent larger. This heterogeneous response holds even after controlling for the national size of firms as well as their age, defining local labor markets using commuting zones rather than counties, or considering different types of monetary policy shock series. Finally, we decompose the wage bill response into changes in employment and changes in firm wage, and find that employment accounts for about 40% of the overall wage bill change.

The heterogeneous response we find across firms suggests that the presence and the observed changes in labor market power over time can affect the transmission of monetary policy to real outcomes. To gauge the aggregate effects of labor market power on monetary policy efficacy, we construct a structural general equilibrium model with heterogeneous oligopsonistic firms and nominal rigidities. The labor market structure builds on [Berger et al. \(2022\)](#). The economy consists of a continuum of local labor markets. Each local labor market contains a finite number of firms that differ along their productivity level. Workers in the economy have idiosyncratic tastes towards different local labor markets and towards different firms, resulting in a finite labor supply elasticity for each firm, which endogenously varies with the wage bill share of the firm in the local market.

The oligopsonistic firms produce a homogeneous good, but set their wages taking into account the labor supply function they face. They compete à la Bertrand and can adjust their wage infrequently. When a firm adjusts its wage, it internalizes its effect on the local wage index, taking as given the wage strategies of its competitors. We study the equilibrium with commitment, which allows us to accommodate rich heterogeneity across firms and local labor markets. The optimal markdown reflects a weighted average of the inverse labor supply elasticities until the next wage adjustment. The labor supply elasticity of the firm moves over time together with its labor market share, even when its nominal wage remains fixed, because of changes in competitors' wages. Since labor supply elasticity is lower for firms with a larger labor market share, large firms set a larger markdown relative to competitive wages. The rest of our model is standard.

Before quantifying the model, we use a special case to show the importance of wage stickiness in driving the heterogeneous response across firms. We consider the limit case where the discount factor is equal to zero, so that optimal wage setting is static. Using this version of the model, we study the effect of a rise in demand, induced by an expansionary monetary policy, on a local labor market consisting of three firms: a low-, medium-, and high-productivity firm. We first show that, absent wage stickiness, all firms adjust their wage by the same magnitude, resulting in a homogeneous response of labor and wage bill despite the heterogeneity in labor market power. That is, when all firms can adjust their wage, their labor market share remains unchanged, they do not change their markdowns, so there is full and homogeneous passthrough from the demand shock to the wage. In other words, absent wage rigidities, the presence of labor market power alone does not change the transmission of monetary policy in this class of models. We then consider the case in which one firm cannot adjust its wage. When that is the case, the wage-bill share of the other two firms goes up, but in a heterogeneous

way. Because the low-productivity firm faces a higher superelasticity of labor supply, its passthrough from a demand shock to wages is higher, and it responds more both in terms of employment and wage bill. Thus, wage stickiness is a key driver of the heterogeneous response across firms.

The stylized model uncovers the two channels through which imperfect competition in the labor market dampens the effect of monetary policy on output. First, oligopsonistic competition reduces the passthrough from prices to wages. This is because firms optimally increase their markdowns when demand goes up. The lower passthrough to wages reduces the overall employment response, and as a result, the output response. This is the *partial passthrough channel*. Second, the heterogeneous response across firms affects aggregate total factor productivity (TFP). An expansionary monetary policy shock increases the employment share of low-monopsony-power firms. These firms are less productive on average, so such a shift reduces aggregate TFP. As a result, oligopsonistic competition dampens the effect of monetary policy on output by moving TFP in the opposite direction. This is the *misallocation channel*.

Solving the full version of the model brings two new challenges relative to standard heterogeneous agent New Keynesian models, such as [Kaplan et al. \(2018\)](#). First, the finite number of firms operating in each local labor market implies that a law of large numbers cannot be invoked to eliminate the idiosyncratic risk of wage stickiness. Second, firms and workers interact in a large number of markets and the equilibrium wages are determined implicitly from firms' mutual best responses. To overcome these challenges, we adapt the sequence-space Jacobian approach ([Auclert et al., 2021](#)) in two ways. First, we extend the perfect foresight assumption from including only aggregate variables to also including the idiosyncratic wage adjustment shocks. Second, we introduce an inner layer where we solve for the equilibrium in each local labor market in isolation. This numerical approach allows us to handle many heterogeneous firms across many local labor markets. The calibrated model consists of approximately one million firms across twenty five thousand labor markets.

We calibrate the model to match key features of the U.S. labor market. Specifically, we target the employment and population share of high-monopsony-power firms, the average HHI across local labor markets, as well as the average markdown. The model is able to replicate the heterogeneous response across low- and high-monopsony-power firms. In response to an expansionary monetary policy shock, the wage bill of low-monopsony-power firms responds by roughly 50 percent more than that of high-monopsony-power firms.

Labor market power affects the transmission of monetary policy through the partial and heterogeneous passthrough of prices to wages by the oligopsonistic firms. To quantify how oligopsonistic competition affects monetary transmission, we consider a counterfactual economy in which firms strive to keep their markdowns constant. This formulation extends the full passthrough benchmark, which would arise without oligopsonistic competition and wage rigidity, to an environment with only wage rigidity. We find that the response of output to a monetary policy shock in the benchmark economy is 12 percent lower than in the full-passthrough counterfactual. Two-thirds of the difference is attributable to the partial passthrough channel and one-third to misallocation.

Finally, we recalibrate the distribution of firms to match the higher level of local concentration observed in U.S. labor markets in 1980. In this more concentrated economy, firms face fewer local

competitors and, on average, hold larger market shares. We find that monetary policy had smaller effects on output but somewhat larger effects on inflation in the 1980s than in the 2010s. This suggests that the decline in local concentration has flattened the Phillips curve over time, accounting for about 15 percent of the flattening estimated by [Blanchard \(2016\)](#) between 1980 and the 2010s. A flatter Phillips curve can either help or hinder the central bank, depending on the policy context: easing becomes less inflationary, while disinflation requires a larger output sacrifice.

Literature review. This paper contributes to empirical and theoretical research on labor market power, firm heterogeneity, and monetary policy.

A recent literature leverages administrative data to estimate the degree of labor market power and the evolution of local labor market concentration over time ([Azar, Marinescu, Steinbaum, and Taska, 2020](#); [Benmelech, Bergman, and Kim, 2022](#); [Rinz, 2022](#); [Chan, Salgado, and Xu, 2023](#)). Our definition of local labor markets follows [Yeh, Macaluso, and Hershbein \(2022\)](#). We contribute to this literature by connecting local labor market concentration and the transmission of monetary policy.

By showing that firms respond differentially to monetary policy shocks according to their local labor market share, our paper relates to the growing literature that studies the role of firm heterogeneity in monetary policy. This literature has emphasized the role of heterogeneity in size and financial constraints ([Gertler and Gilchrist, 1994](#); [Caglio, Darst, and Kalemli-Özcan, 2021](#); [Champagne and Gouin-Bonenfant, 2026](#)), age ([Durante, Ferrando, and Vermeulen, 2022](#); [Cloyne, Ferreira, Froemel, and Surico, 2023](#)), and default risk ([Ottonello and Winberry, 2020](#)). To this list, we add labor market power as a dimension of heterogeneity that modulates firms’ responses to monetary policy.

Modeling firm-side labor market power in general equilibrium is typically done in one of two ways: search and matching frictions, as in [Burdett and Mortensen \(1998\)](#) and [Manning \(2003\)](#), or preference heterogeneity as in [Card, Cardoso, Heining, and Kline \(2018\)](#) and [Lamadon, Mogstad, and Setzler \(2022\)](#). Both of these sources of labor market power have been explicitly recognized by [Robinson \(1933\)](#) who introduced the term “monopsony”. We adopt the approach based on preference heterogeneity, extending the setup of [Berger, Herkenhoff, and Mongey \(2022\)](#) with a nested CES labor supply system to nominal wage stickiness. In the resulting model, heterogeneous firms play a dynamic game with their local competitors on a large number of markets. Solving this model numerically is challenging. We develop a novel application of the sequence-space Jacobian method of [Auclert, Bardóczy, Rognlie, and Straub \(2021\)](#) that scales efficiently to heterogeneous firms and markets.

Finally, our paper is related to the literature on monetary policy with oligopolistic product markets. [Mongey \(2021\)](#) and [Ueda \(2023\)](#) study duopolies, while [Wang and Werning \(2022\)](#) and [Alexander, Han, Kryvtsov, and Tomlin \(2024\)](#) study oligopolies with homogeneous firms. The overarching conclusion of this literature is that imperfect competition in *product markets* increases the efficacy of monetary policy. We show that imperfect competition in *labor markets* leads to the opposite effect—monetary policy is less effective. Across both markets, imperfect competition leads to partial passthrough. But while a partial passthrough from costs to prices amplifies the quantity response in product markets, partial passthrough to wages instead dampens the quantity response. The core of the difference is about which side of the market is subject to nominal rigidities. In product markets, it is the supply

side, while in the labor market, it is the demand side. Our framework also highlights the importance of the misallocation channel which arises with heterogeneous firms. Conceptually, this is a “supply-side channel” of monetary policy that is related to the mechanisms in [Baqae, Farhi, and Sangani \(2024\)](#) and [Meier and Reinelt \(2024\)](#), who show how heterogeneous markup responses to monetary policy across firms affects aggregate TFP.

Layout. The paper proceeds as follows. In section 2, we establish empirically that firms respond differentially to monetary policy shocks depending on the degree of their labor market power. In section 3, we present our heterogeneous agent New Keynesian model with oligopsonistic labor markets. In section 4, we provide intuition using a special case of our model. In section 5, we introduce a numerical method to solve the model and quantify the effect of oligopsonistic competition on the transmission of monetary policy. Section 6 concludes.

2 Empirical Analysis

In this section, we study how low- and high-monopsony-power firms respond to monetary policy shocks. To this end, we combine two administrative data sources from the U.S. Census Bureau. Our main finding is that the wage bill of low-monopsony-power firms is significantly more responsive to monetary policy. We show that this result holds across different specifications. Finally, we decompose the relative wage-bill response into the wage rate and employment responses.

2.1 Data and Descriptive Statistics

Our primary source of data is the Census’ Longitudinal Employer-Household Dynamics (LEHD), which is an employer-employee matched panel dataset constructed from unemployment insurance records. The LEHD contains quarterly data on employment and wages at the worker level, including information on their employer, such as a firm identifier, location of the job, and narrow industry groups. In our analysis, we use quarterly information from 1993:Q1 to 2019:Q4 for all states available. This implies that the number of states increases over time as new ones enter the sample.³

We aggregate LEHD observations to the firm-county-quarter level. Our wage-bill measure includes all earnings paid during the quarter, regardless of the employment-spell duration. Our employment measure excludes worker-firm matches with quarterly earnings below a \$950 wage-bill threshold. This threshold corresponds roughly to three weeks of full-time work at the federal minimum wage, so it removes workers with only minimal attachment to the firm during the quarter. This restriction reduces the influence of transitory matches and produces an employment measure that more accurately

³We have information on 6 states in 1993, including California. By 2000 the number of states for which we observe data grows to 21, and to 23 by 2004. By the end of our sample, we have the following 23 states: Arizona, California, Colorado, Connecticut, Delaware, Kansas, Maine, Maryland, Nevada, New Jersey, New Mexico, North Dakota, Ohio, Oklahoma, Pennsylvania, South Carolina, South Dakota, Tennessee, Utah, Virginia, Washington, Wisconsin, Wyoming. These states account for roughly half of U.S. employment.

captures the firm’s persistent workforce. Our baseline sample contains 207 million firm-county-quarter observations with positive wage bills.

We define a local labor market as the collection of all firms within a subsector (3-digit NAICS) in a county. Then, for each firm in a local labor market, we calculate the share of the labor market wage bill it accounts for. We classify firms into low- and high-monopsony-power categories: a firm is considered to have high monopsony power if it accounts for 10 percent or more of the total wage bill within the local labor market.⁴ All other firms within a labor market are classified as having low monopsony power. Notice that because we aggregate wage bill and employment of firms within subsectors and counties, a firm present in multiple labor markets can be classified as having high monopsony power in one labor market (e.g., a grocery chain being the only supermarket in a small town) but as having low monopsony power in a different one (e.g., the same chain but in a big city).

We combine the data from LEHD with information from the Longitudinal Business Database (LBD), which contains annual information on wage bill and employment for all firms in the private sector from 1978 to 2019. The key advantage of LBD relative to LEHD is its coverage, which spans the entire private business sector for all states. We use LBD to obtain the age of the firm, the levels of employment, and the wage bill at the national and local labor market levels. Furthermore, the geographical information contained in LBD allows us to define a local labor market using Commuting Zones rather than counties, which we use for a robustness analysis. Finally, we combine this dataset with quarterly macroeconomic aggregates (e.g., GDP and unemployment) and a measure of monetary policy shocks obtained from [Jarociński and Karádi \(2020\)](#).

Descriptive Statistics. Table 1 shows summary statistics for a typical quarter in our sample. Firms classified as high-monopsony-power account for about a third of the total wage bill and employment, even though they only account for 6 percent of all firms. These firms operate in markets that are, on average, more concentrated than low-monopsony-power firms. Moreover, there is a larger share of firms having more than 100 employees among high-monopsony-power firms (18.1% versus 1.6%). These firms also tend to be older, with a larger share of firms older than 10 years (81.4% versus 64%).

2.2 Econometric Specification and Results

To measure the impact of monetary shocks across firms, we run a series of lag-augmented local projections ([Jordà, 2005](#); [Montiel Olea and Plagborg-Møller, 2021](#)) of the form

$$\tilde{y}_{i,t+h} - y_{i,t-1} = \delta_{t,h}^s + \beta_h \varepsilon_t^{MP} \times \mathbb{1}[s_{i,t-1} \leq 10\%] + \Gamma_h \mathbf{X}_{i,t-1} + \epsilon_{i,t+h}, \quad \text{for } h = \{0, \dots, 15\} \quad (2.1)$$

where $\tilde{y}_{i,t+h} = \frac{1}{4} \sum_{\tau=0}^3 y_{i,t+h+\tau}$ is a forward-looking moving average of the log of the wage bill for firm i in quarter $t+h$, and $y_{i,t-1}$ is the log of the wage bill in period $t-1$. This simple moving-average local projection specification, which is motivated by the smooth local projection approach of [Barnichon and Brownlees \(2019\)](#), has the advantage that it can be easily implemented with large panel datasets and

⁴The wage-bill share of the firm is the proper measure of labor market power in the model we develop in Section 3. That is, the labor supply elasticity faced by a firm is a decreasing function of its local-labor-market wage-bill share.

Table 1: Summary Statistics by Monopsony Group

	All	Low Monopsony	High Monopsony
Share of firms (%)	—	94.1	5.9
Share of employment (%)	—	61.0	39.0
Share of wage bill (%)	—	56.6	43.4
Local labor market HHI	0.11	0.10	0.42
Share > 100 employees (%)	2.5	1.6	18.1
Share $\geq$ 10 years (%)	64.5	64.0	81.4

Notes: Statistics based on data for 2015:Q1. Firm, employment, and wage-bill shares from LBD. Age and size shares are firm count shares within each monopsony group from LEHD. Local labor market HHI denotes the average local-labor-market Herfindahl-Hirschman index in the markets in which firms operate.

smooths out the jumps of wage bill and employment that are usually observed in firm-level data.

The vector $\mathbf{X}_{i,t}$ contains aggregate controls (CPI inflation, unemployment rate, and GDP interacted with the monopsony-power fixed effect), firm-level controls (four lags of $y_{i,t}$), state-by-subsector-by-monopsony-power fixed effects, and quarter-by-monopsony-power fixed effects to control for any seasonal differences between high- and low-monopsony-power firms.⁵ We also include time-subsector fixed effects $\delta_{t,h}^s$ to control for other aggregate effects that could impact firms' wage bill within an industry. The coefficient β_h captures the response of the wage bill of a firm with low monopsony power (identified by the dummy $\mathbb{1}[s_{i,t-1} \leq 10\%]$) to a monetary policy surprise, ε_t^{MP} , relative to a high-monopsony-power firm.

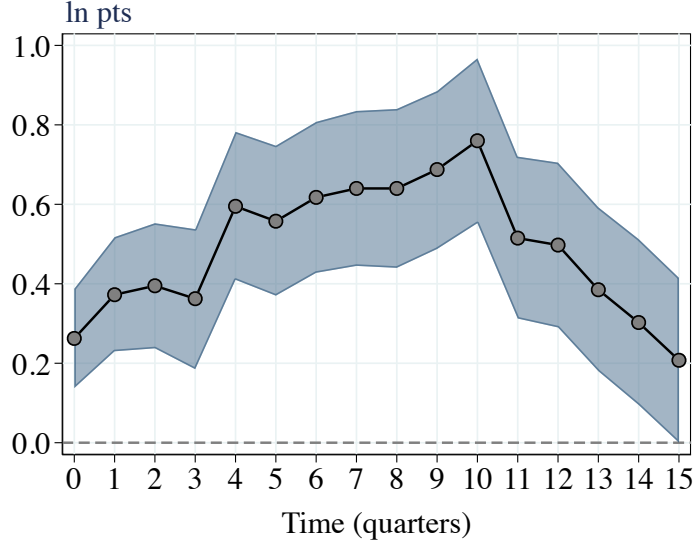
Results. Our baseline results are displayed in Figure 1. The line plots the dynamic response of the log wage bill of low- relative to high-monopsony-power firms to a 25bp expansionary monetary policy shock. The relative response is positive and statistically significant. Specifically, the wage bill increases by 0.6 log points more for low- relative to high-monopsony-power firms on average over the second year following the shock, with a peak effect of about 0.8 log points for the four-quarter window beginning at horizon ten.⁶

To better gauge the economic significance of the result, we also run a local projection for each of the two groups separately, mimicking specification (2.1) but without any time fixed effects and where the monetary policy shock is not interacted with the monopsony group indicator. These results are presented in Appendix Figure C.2. Importantly, both groups respond positively to an expansionary monetary shock: on average over the second year after the shock, the wage-bill response of high-monopsony-power firms is about 2 log points, whereas the response of low-monopsony-power firms is 2.6 log points. On average, over the impulse response horizon, the response of low-monopsony-power

⁵As shown by Montiel Olea and Plagborg-Møller (2021), the lag-augmented specification eliminates the serial-correlation problem in local projection residuals, so standard errors need not be additionally corrected for serial correlation.

⁶For comparison, Appendix Figure C.1 displays the results of a standard local projection where the dependent variable is $y_{i,t+h} - y_{i,t-1}$. The results are similar but noisier than our moving-average specification.

Figure 1: WAGE-BILL RESPONSE TO A MONETARY POLICY SHOCK



Notes: The figure shows the relative response of the firm-level log wage bill to a monetary policy shock of 25 basis points. We plot the coefficient β_h for each horizon h from 0 to 15. The y-axis reports responses in log points: 1 log point corresponds to an approximate 1 percent change. The shaded areas represent 95 percent confidence intervals. Standard errors clustered at the local labor market level.

firms is about 40% larger.⁷

2.3 Robustness

We consider three sets of robustness exercises. First, we assess whether the heterogeneous response we document can be explained by other firm characteristics, in particular firm age and size. Second, we vary the definition of local labor markets and the identification of monetary policy shocks. Third, we examine how our results change as we vary the wage-bill threshold used to classify firms by monopsony power. Figure 2 summarizes the results, reporting the wage-bill response of low-monopsony firms relative to high-monopsony firms, averaged over the second year following the shock.⁸

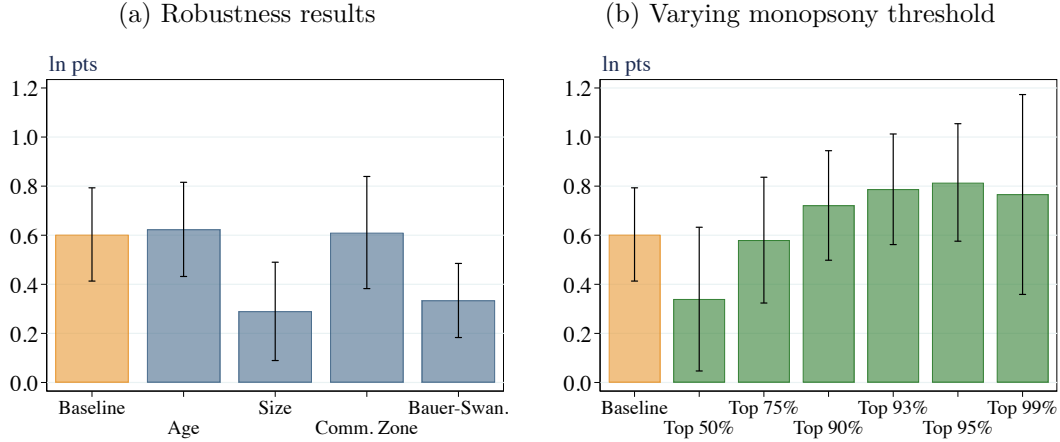
We begin with age and size. As shown in Table 1, low-monopsony firms tend to be smaller and younger than high-monopsony firms, raising the concern that our baseline estimates capture differences along these dimensions rather than differences in labor market power. For example, younger and smaller firms may be more financially constrained, and their relative response to monetary policy may be driven by financial factors rather than labor market power.⁹ To address the age channel, we divide

⁷We note that the overall magnitudes we find are in line with the literature. Champagne and Gouin-Bonenfant (2026) uses administrative Canadian data and find that a 25bp expansionary monetary policy shock increases employment by 1 log point 2 years after the shock. Using aggregate U.S. data, Bauer and Swanson (2023) find that such shock leads to a rise in industrial production of about 1 log point 2 years after the shock (right column of Figure 3 therein). In comparison, the average wage-bill response we estimate at that horizon is about 2.5 log points, and encompasses both the wage rate and employment response. In the next section, we show that employment accounts for about 40% of the wage-bill response.

⁸Appendix Figures C.3 and C.4 show the average response one, three, and four years after the shock.

⁹The evidence on how financial factors shape firms' response to monetary policy is somewhat mixed. Gertler and Gilchrist (1994) find that small firms' sales and inventories are more responsive to monetary policy, and argue that this is largely due to financial factors. Champagne and Gouin-Bonenfant (2026) find that the employment of financially

Figure 2: ALTERNATIVE SPECIFICATIONS



Notes: Panel (a) shows the wage-bill response of low-monopsony-power firms relative to high-monopsony-power firms to a 25-basis-point expansionary monetary policy shock in the second year after the shock, for different specifications. The whiskers represent 95 percent confidence intervals. Panel (b) shows how the relative response varies as we vary the cutoff definition for high-monopsony firms. For these results, low-monopsony-power firms are defined as those below the 50% cutoff.

firms into three age bins (under 5 years, 5 to 10 years, and 10 or more years) and interact the monetary shock with these bins. The second bar of Panel (a) of Figure 2 shows that controlling for age leaves the relative response essentially unchanged. While age is an important determinant of firm cyclical behavior (Pugsley and Şahin, 2019), it does not absorb the effect of relative size within a local labor market.

We proceed analogously for size, dividing firms by lagged employment into four bins: (i) under 5 workers, (ii) 5 to 10 workers, (iii) 10 to 20 workers, and (iv) above 20 workers. The results, presented in the third bar of Panel (a) of Figure 2, show that controlling for size attenuates the estimate, but a sizable and statistically significant difference remains: low-monopsony firms still respond about 0.25 log points more than high-monopsony firms. The attenuation may reflect financial frictions, or that firm size is closely tied to labor market share and absorbs part of the underlying monopsony variation. Overall, these results indicate that differences in firm age and size may account for part of the heterogeneous response, but do not overturn the role of monopsony power.

We next consider the definition of the local labor market. Our baseline defines a local labor market as a county-subsector pair, following Yeh et al. (2022). A natural concern is that labor markets are better defined at the occupation-commuting-zone level. The LEHD does not contain occupational information, and our sample only covers a subset of states, so we cannot observe the full set of firms operating in commuting zones that span multiple states at a quarterly frequency. Regarding occupation, Handwerker and Dey (2024) show that measures of labor market concentration based on industries and occupations are highly correlated across geographic areas, alleviating concerns that focusing on industries rather than occupations introduces substantial measurement error. To address the geographic dimension, we re-estimate our analysis defining local labor markets at the

constrained firms is more responsive to monetary policy. Ottonello and Winberry (2020), on the other hand, find that the investment of financially constrained firms—those with higher debt burdens and lower “distance to default”—is *less* responsive to monetary policy.

industry–commuting-zone level using the nationwide coverage of the LBD which we use to construct firms’ labor market shares and define firms monopsony power. We then merge this measure to our LEHD sample. A caveat is that this classification can only be implemented at the annual frequency, rather than quarterly as in our baseline. The fourth bar of Panel (a) of Figure 2 presents the results: low-monopsony firms remain significantly more responsive to monetary policy shocks, with a slightly larger coefficient than in the baseline specification. Overall, these results indicate that our main findings are robust to alternative definitions of the local labor market.

As an additional robustness check, we consider an alternative measure of monetary policy shocks. Our baseline uses the high-frequency shocks of Jarociński and Karádi (2020), while here we replace them with the shocks constructed by Bauer and Swanson (2023), which account for the Fed information effect differently. Re-estimating our baseline specification with this alternative measure yields qualitatively similar results: low-monopsony firms remain significantly more responsive to monetary policy shocks, although the magnitude of the differential response is smaller, around 0.3 log points compared to 0.6 in the baseline. These results indicate that our findings are robust to alternative measures of monetary policy shocks.

Finally, we examine how our results vary with the threshold used to define high-monopsony-power firms. To this end, we construct alternative cutoffs based on percentiles of the distribution of wage-bill shares across all firms. Specifically, we fix the group of low-monopsony firms as those below the 50th percentile of the nationwide distribution, and define high-monopsony firms using progressively higher thresholds, ranging from the 50th to the 99th percentile. Panel (b) of Figure 2 reports the resulting estimates, focusing on the relative wage-bill response in the second year following the shock.

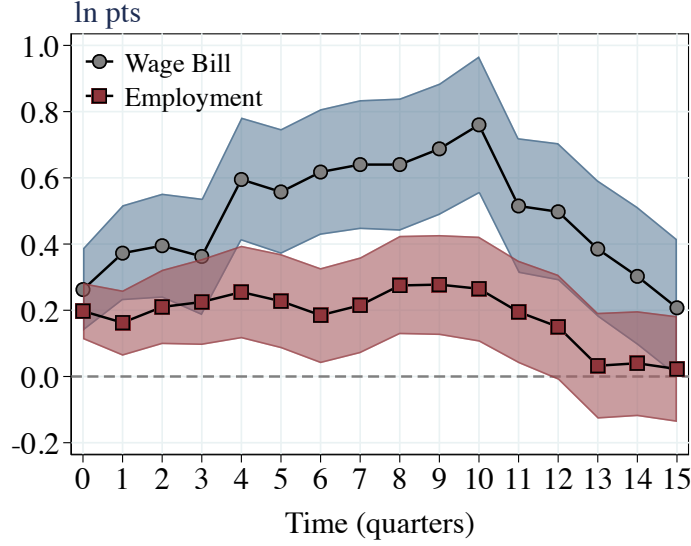
Consistent with our baseline results, low-monopsony firms are more responsive to monetary policy shocks for all cutoff definitions. Moreover, the differential response increases with the threshold: firms in the bottom half of the wage-bill share distribution respond about 0.35 log points more than firms above the 50th percentile, and about 0.8 log points more than firms above the 95th percentile.¹⁰ This pattern is consistent with the mechanism in our model, whereby firms’ labor market power is continuous rather than binary.

2.4 Decomposing the wage-bill response

To better understand the sources of the wage-bill response, we examine the contribution of employment and wages separately. Figure 3 reports the dynamic response of employment alongside that of the wage bill. The employment response is positive and statistically larger for low-monopsony-power firms, but smaller in magnitude than the wage-bill response. In the second year after the shock, employment growth at low-monopsony firms is about 0.25 log points higher than at high-monopsony firms, accounting for 42% of the overall differential in the wage bill. The remaining 58% is driven by wage growth. This pattern suggests that differences in wage adjustment play a central role in driving the heterogeneous response across firms, consistent with evidence that firms with lower labor market

¹⁰The difference between the baseline specification and the fifth bar in Panel (b) of Figure 2, which uses the same cutoff, lies in the definition of the low-monopsony group. In the baseline, this group includes all firms with wage-bill shares below 10 percent, whereas in the alternative specification it consists of firms below the median share.

Figure 3: EMPLOYMENT RESPONSE TO A MONETARY POLICY SHOCK



Notes: The figure shows the relative response of the firm-level log employment (red line with squares) and log wage bill (blue line with circles). We plot the coefficient β_h for each horizon h from 0 to 15. The shaded areas represent 95 percent confidence intervals. Standard errors clustered at the local labor market level.

power exhibit stronger wage responses to shocks (Chan et al., 2023).

3 Model

In this section, we present our oligopsonistic New Keynesian model with heterogeneous firms. To model imperfect competition, we build on the approach developed in the trade literature by Atkeson and Burstein (2008) and adapted to labor markets by Berger, Herkenhoff, and Mongey (2022). Firms are atomistic in national markets but large in their local labor market. This type of model is well-suited to account for the empirical evidence we presented in Section 2 because it provides a theory of market shares and passthrough at the firm level within local labor markets.

The distinguishing feature of our model is wage stickiness. As we show, wage stickiness is necessary to replicate the heterogeneous passthrough of monetary policy shocks. However, wage stickiness breaks the block recursivity property that makes most models of this type tractable.¹¹ Related models with nominal rigidities include Mongey (2021), who focuses on duopolies, and Wang and Werning (2022), who focus on symmetric firms. Relative to these seminal papers, we allow for any degree of firm heterogeneity in terms of productivity and the number of local competitors, which enables our model to match the cross-sectional moments that shape our quantitative results. The key assumption that makes our model tractable is that firms formulate their wage strategies in sequence space and can credibly commit to them. Given this assumption, we can leverage the sequence-space Jacobian

¹¹For example, Berger et al. (2022) prove that their model is block recursive, meaning that local labor market equilibria can be characterized independently of aggregates. Wage stickiness breaks block recursivity, because it forces firms to play a dynamic game in which aggregate fluctuations are payoff relevant.

method to solve for Nash equilibria in thousands of local markets inhabited by hundreds of thousands of heterogeneous firms, in a way that depends on monetary policy and is consistent with general equilibrium.

Environment. Time is discrete and each period corresponds to a quarter. The economy consists of a representative household, a continuum of firms, and a central bank. There are two types of firms: producers and retailers. Producers inhabit a continuum of local labor markets $j \in [0, 1]$, each with a finite number of firms indexed by $i \in \{1, 2, \dots, M_j\}$. They hire labor from households and produce a homogeneous intermediate good, which they sell to retailers in national markets. Retailers differentiate the intermediate good into imperfectly substitutable varieties indexed by $k \in [0, 1]$ which they sell to the household. Producers have monopsony power in the labor market. Retailers have monopoly power in the final-goods market. We separate producers and retailers, and assume that retailers are not affected by the boundaries of local labor markets, for tractability.

3.1 Households

The representative household chooses sequences of consumption $\{c_{kt}\}$ from retailer k , labor supply $\{n_{ijt}\}$ to producer ij , and nominal bonds $\{B_t\}$ to maximize the present value of expected utility. Given initial wealth B_0 , the household solves

$$\max_{c_{kt}, n_{ijt}, B_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t), \quad (3.1)$$

where the aggregate consumption and labor supply indices are given by

$$C_t = \left(\int_0^1 c_{kt}^{\frac{\epsilon-1}{\epsilon}} dk \right)^{\frac{\epsilon}{\epsilon-1}}, \quad N_t = \left(\int_0^1 n_{jt}^{\frac{\zeta+1}{\zeta}} dj \right)^{\frac{\zeta}{\zeta+1}}, \quad n_{jt} = \left(\sum_{i=1}^{M_j} n_{ijt}^{\frac{\eta+1}{\eta}} \right)^{\frac{\eta}{\eta+1}}, \quad (3.2)$$

with elasticities $\epsilon > 1$, and $\eta > \zeta > 0$. The ordering of the labor supply elasticities implies that the representative household is more willing to substitute labor within a local labor market than across markets. Utility maximization is subject to the following budget constraint every period

$$\int_0^1 p_{kt} c_{kt} dk + B_t = (1 + i_{t-1}) B_{t-1} + \int_0^1 \left(\sum_{i=1}^{M_j} W_{ijt} n_{ijt} \right) dj + \Pi_t, \quad (3.3)$$

where p_{kt} is the price of final good k , i_{t-1} is the nominal return on bonds between periods $t-1$ and t , W_{ijt} is the nominal wage of producer i in local labor market j , and Π_t is aggregate profits.

Optimality conditions. Demand for goods is summarized by an aggregate Euler equation and retailer-specific demand curves

$$U_C(C_t, N_t) = \beta \mathbb{E}_t \left[\frac{1 + i_t}{\pi_{t+1}} U_C(C_{t+1}, N_{t+1}) \right], \quad c_{kt} = \left(\frac{p_{kt}}{P_t} \right)^{-\epsilon} C_t, \quad (3.4)$$

where the aggregate price index and inflation can be expressed as

$$P_t = \left(\int_0^1 p_{kt}^{1-\epsilon} dk \right)^{\frac{1}{1-\epsilon}}, \quad \pi_t = \frac{P_t}{P_{t-1}}. \quad (3.5)$$

Similarly, aggregate and firm-specific labor supply are given by

$$\frac{W_t}{P_t} = -\frac{U_N(C_t, N_t)}{U_C(C_t, N_t)}, \quad n_{ijt} = \left(\frac{W_{ijt}}{W_{jt}} \right)^\eta \left(\frac{W_{jt}}{W_t} \right)^\zeta N_t, \quad (3.6)$$

where local and aggregate wage indices are

$$W_t = \left(\int_0^1 W_{jt}^{1+\zeta} dj \right)^{\frac{1}{1+\zeta}}, \quad W_{jt} = \left(\sum_{i=1}^{M_j} W_{ijt}^{1+\eta} \right)^{\frac{1}{1+\eta}}. \quad (3.7)$$

See Appendix A.1 for derivations.

Labor market power. The CES labor supply system described by equations (3.6) and (3.7) implies that, as long as the elasticities η and ζ are finite, producers will have labor market power. Producer ij may offer a lower wage than its local competitors ($W_{ijt} < W_{jt}$) and national competitors ($W_{ijt} < W_t$) and it will still attract workers. Finite labor supply elasticities capture, in reduced form, the deeper reasons why firms have labor market power: mobility costs, search costs, specialized human capital, and heterogeneous preferences over non-wage amenities (Berger et al., 2022).

3.2 Retailers

There is a unit measure of retailers $k \in [0, 1]$ who buy a homogeneous intermediate good at price M_t and turn it into differentiated final goods c_{kt} which they sell to households at price p_{kt} . Since varieties are imperfect substitutes, retailers have monopoly power and set their own prices. There is price stickiness à la Calvo: retailers can adjust their price with probability $1 - \theta_p \in (0, 1)$. The probability of adjustment is i.i.d. across periods and firms.

A retailer that cannot adjust its price has no decision to make. It produces just enough to satisfy household demand. A retailer that can reset its price will maximize the present value of expected profits until the next price adjustment

$$\max_{p_{kt}^*, \{c_{kt+\tau}, y_{kt+\tau}\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \theta_p^\tau R_{t,t+\tau} (p_{kt}^* c_{kt+\tau} - M_{t+\tau} y_{kt+\tau}) \quad (3.8)$$

subject to the demand curve

$$c_{kt+\tau} = \left(\frac{p_{kt}^*}{P_{t+\tau}} \right)^{-\epsilon} C_{t+\tau}, \quad (3.9)$$

where $R_{t,t+\tau}$ is the nominal stochastic discount factor between periods t and $t + \tau$,

$$R_{t,t+\tau} = \beta^\tau \frac{U_C(C_{t+\tau}, N_{t+\tau})}{U_C(C_t, N_t)} \frac{P_t}{P_{t+\tau}}. \quad (3.10)$$

Price setting equation. Although the retailer's problem is standard, it is useful to review it due to its similarity to the novel problem of the oligopsonistic producers. Price adjusters are symmetric and set the same price

$$p_t^* = \frac{\epsilon}{\epsilon - 1} \frac{\mathbb{E}_t \sum_{\tau=0}^{\infty} \theta_p^\tau R_{t,t+\tau} P_{t+\tau}^\epsilon C_{t+\tau} M_{t+\tau}}{\mathbb{E}_t \sum_{\tau=0}^{\infty} \theta_p^\tau R_{t,t+\tau} P_{t+\tau}^\epsilon C_{t+\tau}}. \quad (3.11)$$

Equation (3.11) shows that all retailers have the same market power, captured by a *desired markup* that is constant and equal to $\epsilon/(\epsilon - 1)$. The firm sets p_t^* to achieve this markup over the expected average marginal cost until the next price adjustment. The marginal cost τ periods ahead is discounted with the nominal discount factor $R_{t,t+\tau}$, with the probability θ_p^τ that the price is still in effect, and weighted by expected sales which are proportional to $P_{t+\tau}^\epsilon C_{t+\tau}$.

Log-linearizing the first order condition (3.11) in conjunction with the price index (3.5) yields the textbook New Keynesian Phillips curve

$$\hat{\pi}_t = \frac{(1 - \theta_p)(1 - \beta\theta_p)}{\theta_p} \hat{m}_t + \beta \hat{\pi}_{t+1}, \quad (3.12)$$

where $\hat{\pi}_t$ and $\hat{m}_t$ denote inflation and real marginal cost in log-deviations from the zero-inflation steady state. See Appendix A.2 for derivations.

3.3 Producers

The producers' problem is ultimately similar to the textbook price setting problem of the retailers, with the key difference that it allows for heterogeneous labor market power across firms. There is a conceptual difference as well, namely that firms play a noncooperative dynamic game with their local competitors. We describe the game under full commitment and argue that the resulting wage strategies are time consistent.

There is a continuum of producers who are ex-ante heterogeneous in two dimensions. First, each firm operates in a specific local labor market $j \in [0, 1]$ inhabited by a finite number of firms, M_j , which varies across local labor markets. Second, firms differ in their productivity, z_{ij} , which is drawn from a location-invariant distribution $F(z)$ once and for all. In addition, there is Calvo-style nominal wage rigidity: producers can reset their wage with probability $1 - \theta_w \in (0, 1)$ that is i.i.d. over time and across firms. Thus, firms also differ ex post in their histories of Calvo shocks $\{\Theta_{ijt}\}_{t=0}^{\infty}$, where Θ_{ijt} equals one when firm i in sector j can adjust its wage at time t , and zero otherwise.

This market structure implies that producers are infinitesimal with respect to the macroeconomy but non-atomistic in their local labor market. Accordingly, firms take aggregate outcomes as given while engaging with their local competitors in a dynamic game to determine local labor market outcomes. Specifically, we assume that firms compete for workers à la Bertrand with perfect commitment. Let s_t denote the state of the world in period t and s^t the history of states, $s^t = \{s_0, \dots, s_t\}$. The state

s_t consists of a monetary policy shock ϵ_t as well as the Calvo shocks for all firms $\{\{\Theta_{ijt}\}_{i=1}^{M_j}\}_{j \in [0,1]}$. In period 0, all firms commit to a (nominal) wage strategy $W_{ij}(s^t)$ that is a best response to the wage strategies of other firms as well as to the aggregate outcomes. As is standard in New Keynesian models, all firms are required to accommodate labor supply at the prevailing wages.

Formally, firm i in market j solves

$$\max_{\{W_{ij}(s^t), n_{ij}(s^t), W_j(s^t)\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \sum_{s^t} R(s^t) \left[M(s^t) z_{ij} n_{ij}(s^t) - n_{ij}(s^t) W_{ij}(s^t) \right], \quad (3.13)$$

$$\text{s.t.} \quad n_{ij}(s^t) = \left(\frac{W_{ij}(s^t)}{W_j(s^t)} \right)^\eta \left(\frac{W_j(s^t)}{W(s^t)} \right)^\zeta N(s^t), \quad (3.14)$$

$$W_j(s^t) = \left[W_{ij}(s^t)^{1+\eta} + \sum_{i' \neq i} W_{i'j}(s^t)^{1+\eta} \right]^{\frac{1}{1+\eta}}, \quad (3.15)$$

$$W_{ij}(s^t) = W_{ij}(s^{t-1}) \quad \text{if } \Theta_{ijt} = 0. \quad (3.16)$$

The firm sets its wage $W_{ij}(s^t)$ to maximize the present discounted value of future profits, taking into account the labor supply curve (3.14). In doing so, it considers its ability to move the local wage index $W_j(s^t)$, given the wage strategies of its local competitors $W_{i'j}(s^t)$, as shown by (3.15). Finally, (3.16) ensures that the wage strategy is implementable given the Calvo shocks.

Separability across wage spells. Next, we argue that the optimal wage strategy may be solved for in a piece-wise fashion, considering wage spells in isolation.

DEFINITION 1. A wage spell $\mathcal{S}_{ijt}$ is a set of states $s^{t+\tau}$ with $\tau \geq 0$ such that: (i) the history of states is equivalent up to time t , (ii) the firm gets to adjust its wage at time t , $\Theta_{ijt} = 1$, (iii) the firm is stuck with its time- t wage until time $t + \tau$, $\sum_{\tilde{\tau}=0}^{\tau} \Theta_{ijt+\tilde{\tau}} = 1$.

Given this definition, we can partition the firm's problem into wage spells and establish separability between wage spells, allowing us to write the wage setting problem as

$$\max_{\{W_{ijt}^*, n_{ijt+\tau}, W_{jt+\tau}\}} \mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} \left[M_{t+\tau} z_{ij} n_{ijt+\tau} - n_{ijt+\tau} W_{ijt}^* \right], \quad (3.17)$$

$$\text{s.t.} \quad n_{ijt+\tau} = \left(\frac{W_{ijt}^*}{W_{jt+\tau}} \right)^\eta \left(\frac{W_{jt+\tau}}{W_{t+\tau}} \right)^\zeta N_{t+\tau}, \quad (3.18)$$

$$W_{jt+\tau} = \left[(W_{ijt}^*)^{1+\eta} + \sum_{i' \neq i} W_{i'jt+\tau}^{1+\eta} \right]^{\frac{1}{1+\eta}}. \quad (3.19)$$

The derivation, given in Appendix A.3, hinges on our assumption of full commitment. Without commitment, firms would have to consider the off-equilibrium effects that their own wage choice today has on their competitors' future wage choices, leading to dependence between wage spells.¹² At the notational level, the only difference between the objective functions of retailers and producers is that

¹²We discuss the assumption of commitment in more detail in Section 3.6.

retailers only care about their own Calvo shocks and hence the expectations in (3.8) are straightforward to write out without defining price spells.

Wage setting equation. The optimal reset wage, which we formally derive in Appendix A.3, is given by

$$W_{ijt}^* = \frac{\mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} n_{ijt+\tau} [\eta - (\eta - \zeta) s_{ijt+\tau}] M_{t+\tau} z_{ij}}{\mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} n_{ijt+\tau} [1 + \eta - (\eta - \zeta) s_{ijt+\tau}]}, \quad (3.20)$$

where $s_{ijt} \equiv W_{ijt} n_{ijt} / W_{jt} n_{jt}$ is the wage-bill share of firm i in market j at time t . To build intuition, let ϵ_{ijt} denote the labor supply elasticity faced by firm i in market j at time t . Under Bertrand competition and the nested CES structure, this elasticity is

$$\epsilon_{ijt} = \eta - (\eta - \zeta) s_{ijt},$$

a weighted average of the within-market elasticity η and the cross-market elasticity ζ .¹³ Firms with negligible market share compete only locally for workers and therefore face elasticity η . By contrast, a firm that employs all workers in the local labor market competes more broadly for workers across markets and faces the lower elasticity ζ .

If firms could adjust wages period-by-period, the optimal wage would implement the markdown

$$\mu_{ijt} \equiv \frac{1 + \epsilon_{ijt}}{\epsilon_{ijt}} = \frac{1 + \eta - (\eta - \zeta) s_{ijt}}{\eta - (\eta - \zeta) s_{ijt}}. \quad (3.21)$$

Equation (3.20) shows that under wage rigidity, firms instead choose a reset wage to balance expected future markdowns over the duration of the wage spell. Unlike the standard Calvo pricing problem, the desired markdown cannot be factored outside the expectations because it varies over time with the firm's wage-bill share.

Given $\eta > \zeta > 0$, firms with larger wage-bill shares face lower labor supply elasticities and therefore set lower wages relative to the marginal revenue product of labor. In our model, differences in wage-bill shares arise from heterogeneity in productivity and local market structure. Equation (3.21) therefore provides a sufficient statistic for labor market power in wage setting: a firm's wage-bill share summarizes the extent of its monopsony power both within the local labor market and nationally. This observation motivates our empirical classification of firms by local wage-bill share.

3.4 Central Bank

The central bank sets the nominal rate according to a standard Taylor rule

$$i_t = i_{ss} + \phi_{\pi}(\pi_t - 1) + \varepsilon_t. \quad (3.22)$$

¹³Under Cournot competition, as in Berger et al. (2022), the relevant elasticity becomes the harmonic mean

$$\epsilon_{ijt}^{\text{Cournot}} = \left[\frac{1}{\eta} - \left(\frac{1}{\eta} - \frac{1}{\zeta} \right) s_{ijt} \right]^{-1}.$$

We focus on Bertrand competition because, in environments with wage rigidity, it is more natural to model firms as choosing wages rather than employment levels.

The ex-post real return on nominal bonds satisfies the Fisher equation

$$1 + r_t = \frac{1 + i_{t-1}}{\pi_t}. \quad (3.23)$$

3.5 Equilibrium

Given a sequence of monetary policy shocks $\{\varepsilon_t\}$, sequences of Calvo shocks for each producer $\{\Theta_{ijt}\}$, and initial wealth B_{-1} , the equilibrium is a sequence of allocations $\{B_t, C_t, N_t, n_{jt}, n_{ijt}\}$ and prices $\{i_t, r_t, \pi_t, w_{ijt}^*, w_{ijt}, w_{jt}, w_t, m_t\}$ such that

1. Households and producers optimize.
2. Local labor markets are in Nash equilibrium with real wages and wage indices satisfying (3.7).¹⁴
3. Inflation satisfies the log-linearized Phillips curve (3.12).¹⁵
4. The nominal rate satisfies the Taylor rule and the real rate satisfies the Fisher equation.
5. Goods and bond markets clear:

$$Y_t \equiv \int_0^1 \sum_{i=1}^{M_j} z_{ij} n_{ijt} dj = C_t, \quad B_t = 0. \quad (3.24)$$

3.6 Discussion

We now discuss some of the key assumptions of our model.

First, our equilibrium concept assumes that firms have full commitment when setting their wage-path strategies. This assumption enables us to solve the model at the scale that is required to match the empirical evidence. An alternative equilibrium concept that we do not consider is the Markov perfect equilibrium (MPE). [Reinganum and Stokey \(1985\)](#) note that the two approaches correspond to two extreme assumptions about players' ability to commit to future actions. Using path strategies, like we do in this paper, corresponds to the assumption that firms can commit for their entire planning horizon. Using decision rule strategies, as in [Mongey \(2021\)](#) and [Wang and Werning \(2022\)](#), corresponds to the assumption that no commitment is possible at all.¹⁶

A key difference between our equilibrium concept and the MPE is in the level of markdowns in the deterministic steady state. In our setup, steady-state markdowns coincide with those in the flexible-wage equilibrium. In MPE, steady-state markdowns are higher than in the flexible-wage equilibrium. The reason is that a firm internalizes how raising its markdown in the current period can lead to a

¹⁴The wage indices are stated in terms of nominal wages, but they apply to real wages $w_t \equiv W_t/P_t$ as well.

¹⁵Working directly with the log-linearized Phillips curve lets us omit the second-order effects of price dispersion.

¹⁶We note that in our setting, the full-commitment equilibrium does not lead to time-inconsistent solutions as is often the case. This is because a firm is atomistic with respect to the aggregate economy and it takes the wage paths of its competitors as given.

future increase in the markdowns of its competitors.¹⁷ Since the focus of this paper is the transmission of monetary policy with oligopsonistic competition, the concern regarding the choice of equilibrium is not about the steady-state level but rather about the response of the economy to monetary policy. This concern is attenuated by the finding of Wang and Werning (2022), who study the transmission of monetary policy with oligopolistic competition. They show that while the steady-state allocations under MPE and under full-commitment are significantly different, the difference in the response of the economy to a monetary policy shock is minuscule.¹⁸

Second, our model assumes that each local labor market is populated by a finite number of oligopsonistic firms, and that there is constant elasticity of substitution between firms within the local labor market. As we show in the next section, the ability of the model to match the heterogeneous response observed in the data hinges on optimal markdowns being an increasing and convex function of the relative market share of the firm. Such markdown behavior arises with a nested-CES structure with oligopsonistic competition in the inner nest akin to that in Atkeson and Burstein (2008) and Berger et al. (2022), as we adopt in this paper. Alternatively, we could have assumed that there is a continuum of firms within each local labor market but that the elasticity of substitution across these firms is not constant, e.g., Kimball (1995). As long as optimal markdowns remain increasing and convex in the relative market share of the firm, such a change will not affect the qualitative response of the economy to a monetary policy shock. We opted for the nested-CES with oligopsonistic competition structure as it provides a cleaner mapping between model and data. This is because it allows us to: (i) classify firms in the model into low- and high-monopsony-power groups in the same way as we do in the data, (ii) match the HHI distribution observed in the data.

Third, our model assumes that both wages and prices are subject to nominal rigidities. Having two sources of nominal rigidities may seem redundant, but that is not the case. Both sources of nominal rigidity help the model to match key features of the data. First, as we show in the next section, nominal wage rigidities are necessary to obtain the heterogeneous response of firms to monetary policy observed in the data. Second, nominal price rigidities are necessary to match the *average* wage bill response. If prices were flexible, an expansionary demand shock would cause both real wages and employment to fall. With nominal price rigidities, an expansionary monetary policy can lead to an increase in both wages and employment.

¹⁷Competitors' future markdowns may rise for two reasons. First, if the firm is still stuck at its current wage while a competitor resets theirs, strategic complementarities imply that the competitor will optimally choose a higher markdown. Second, even after the firm resets its own wage, that wage can continue to influence future markdowns—through competitors who adjusted their wage during the firm's previous wage spell and remain locked into that level.

¹⁸Wang and Werning (2022)'s measure of monetary non-neutrality is the half-life of the price-level response to a permanent monetary shock. They report that the difference in the half-life under MPE and full commitment is less than 3 percent in all of their specifications. To put that in context, moving from monopolistic to oligopolistic competition increases the half-life by 40 percent in their framework.

4 Monetary Policy Transmission with Oligopsonistic Competition

We next study the transmission of monetary policy in our model. To make the analysis transparent, we consider an analytically tractable special case and characterize the behavior of a local labor market in partial equilibrium. This exercise serves three purposes. First, it shows that nominal wage rigidities are necessary for the model to generate the heterogeneous effects of monetary policy across low- and high-monopsony-power firms observed in the data. Second, it illustrates how the nested-CES structure with oligopsonistic competition naturally gives rise to greater responsiveness among low-monopsony-power firms. Third, it highlights the two channels through which oligopsonistic competition dampens the aggregate effect of monetary policy on output: (i) partial passthrough and (ii) misallocation.

We focus on a local labor market with three firms that differ in productivity levels, $z_1 < z_2 < z_3$. We assume myopic households ($\beta = 0$), so the wage-setting decision is static. Under these assumptions, the optimal wage-setting condition simplifies to

$$W_{it}^* = \frac{1}{\mu_{it}} z_i M_t, \quad (4.1)$$

where M_t is the price of the good sold by the oligopsonistic firms, which they take as given. μ_{it} is the desired markdown of firm i at time t , which is given by

$$\mu_{it} = \frac{1 + \eta - (\eta - \zeta)s_{it}}{\eta - (\eta - \zeta)s_{it}}, \quad (4.2)$$

with s_{it} denoting the firm's wage-bill share in the local labor market. In the full model, the transmission of monetary policy to employment and wages depends on how each local market responds to a change in M_t . Accordingly, we analyze the response of a local labor market to an increase in M_t .

Total differentiation of the wage-setting equation yields

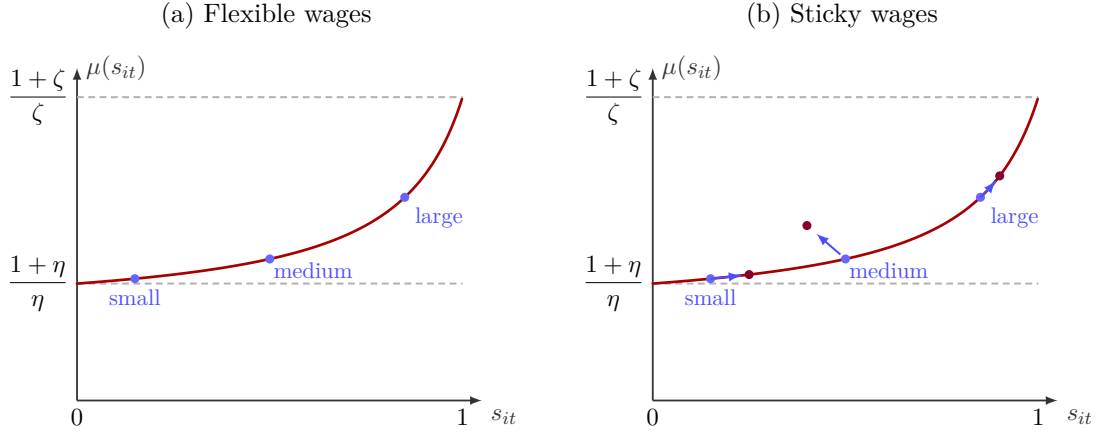
$$\frac{d \ln W_{it}^*}{d \ln M_t} = 1 - \frac{d \ln \mu_{it}}{d \ln M_t}. \quad (4.3)$$

Absent a change in the optimal markdown, the elasticity of wage response to the sales price is unitary. If the markdown optimally rises in response to an increase in the price, there is only partial passthrough: wages adjust less than one-for-one. As equation (4.2) shows, the optimal markdown depends on each firm's wage-bill share, which in turn depends not only on its own wage but also on the wages of competitors in the local labor market. To understand this relationship, we next solve for the equilibrium in the local labor market and highlight the key ingredients that allow the model to replicate the heterogeneous responses across firms observed in the micro data.

4.1 The Heterogeneous Passthrough of Monetary Policy

We begin by analyzing the equilibrium in the local labor market when wages are fully flexible. Consider a rise in the intermediate input price at time $t = 1$. Let $\Delta \ln x_i$ denote the log difference between any variable x of firm i at time 1 and its initial level at time 0. The following proposition characterizes the resulting local labor market equilibrium.

Figure 4: Heterogeneous Response to Monetary Policy in Special Case



Notes: The panels illustrate how markdowns and wage-bill shares respond to an increase in the intermediate good's price under two scenarios. In Panel A (flexible wages), all firms fully pass through the price increase, leaving wage-bill shares and markdowns unchanged. In Panel B (rigid wages), the *medium* firm's nominal wage is fixed, causing its markdown to rise. In equilibrium, the small and large firms expand employment. Because the markdown function is convex, the large firm's markdown rises more than the small firm's, so its wage-bill share increases by less.

PROPOSITION 1. *Consider an increase in the intermediate input price at $t = 1$, so that $M_1 > M_0$. If all firms can adjust their wages, there is a homogeneous response of wages and employment across firms. That is, $\Delta \ln W_i = \Delta \ln W_{i'}$ for all i and i' , and $\Delta \ln s_i = 0$ for all i .*

All formal proofs are provided in Appendix A.4. Proposition 1 implies that, in the absence of wage rigidities, both wage-bill and employment responses are homogeneous across firms. This homogeneity arises despite differences in steady-state markdowns across firms.

Panel A of Figure 4 illustrates the intuition behind the homogeneous passthrough result. The red line plots the optimal markdown as a function of the wage-bill share. Suppose all firms fully pass through the rise in the good's price to their wages. In that case, all wages rise proportionally, and each firm's wage-bill share at time $t = 1$ remains identical to its share at $t = 0$. Because wage-bill shares are unchanged, so are the desired markdowns. Hence, by equation (4.3), our conjecture that full passthrough is the equilibrium outcome is confirmed.

Two key modeling assumptions underlie the homogeneous passthrough result. First, in our framework, the wage-bill share is a sufficient statistic for the optimal markdown. Second, wage-bill shares are homogeneous of degree zero. These assumptions are not specific to the nested-CES preferences we use for the disutility of labor and also hold under alternative formulations such as Kimball (1995) preferences.

We now turn to the equilibrium with nominal wage rigidities. Suppose that one of the three firms cannot adjust its nominal wage. The following proposition shows that in this case, the response becomes heterogeneous across the two firms that can adjust their wages. In particular, both the wage and employment of the smaller firm respond more strongly.

PROPOSITION 2. *Suppose that firm i cannot adjust its wage, so that $\Delta \ln W_i = 0$. Then, among the firms that can adjust their wages, the smaller one expands its wage-bill share by more. That is,*

$\Delta \ln s_j < \Delta \ln s_{j'}$ for $j, j' \neq i$ and $z_j > z_{j'}$.

To build intuition for Proposition 2, suppose that the medium firm’s nominal wage is rigid. Because its wage remains fixed when the price of the good rises, its markdown increases, shifting employment toward the flexible firms. The key question is whether the flexible firms adjust their wages symmetrically. The answer depends on the curvature of the markdown function in equation (4.2), which is governed by the superelasticity of labor supply. When the superelasticity decreases with the wage-bill share, the markdown function is convex, implying that larger firms face a less elastic labor supply than smaller firms.

Panel B of Figure 4 illustrates this mechanism. When the medium firm’s wage is rigid, both the small and large firms expand employment. Because the markdown function is convex, an equal proportional increase in wage-bill shares induces a larger rise in the desired markdown for the relatively larger firm. The smaller firm therefore raises its wage relatively more, while the larger firm’s passthrough is dampened. This stronger wage response also translates into a larger employment increase for the smaller firm. The heterogeneous transmission of monetary policy consistent with our micro evidence thus arises whenever the superelasticity of labor supply decreases with firm size—a condition that holds under the nested-CES structure with $\eta > \zeta$ that we assume.

4.2 Oligopsonistic Competition and the Effectiveness of Monetary Policy

Having established the mechanism underlying the heterogeneous response across firms, we turn to studying the aggregate effect of oligopsonistic competition on the transmission of monetary policy. There are two channels through which oligopsonistic competition dampens the response of employment and output to an expansionary monetary shock: a partial passthrough channel and a misallocation channel.

The first channel operates through incomplete passthrough of prices to wages. With nominal rigidities, the passthrough from the price of goods sold, M_t , to wages is below one. Part of this limited adjustment is mechanical—firms with fixed wages exhibit zero passthrough—but this component is independent of the degree of competition. Oligopsonistic competition further reduces passthrough because firms able to adjust their wages choose to raise their markdowns rather than fully transmit higher prices to wages. When average markdowns rise, the passthrough from prices to wages is incomplete, so nominal shocks translate less into real wage adjustments and employment growth. In effect, oligopsonistic firms absorb part of the monetary expansion by widening their markdowns, weakening the overall sensitivity of labor demand to monetary policy.

The degree of competition in the market directly shapes the extent of passthrough. As shown in Figure 4, the markdown is a convex function of the wage-bill share, implying that passthrough from prices to wages declines with firm size. Consequently, as the number of firms in a local labor market increases, the average passthrough rises, reducing the strength of the partial-passthrough channel.

The second channel through which oligopsonistic competition dampens the effect of monetary policy is the misallocation channel. Recall that the oligopsonistic firms produce a homogeneous inter-

mediate good. To a first-order approximation, aggregate production in the economy is given by¹⁹

$$Y_t \approx \int_0^1 \sum_{i=1}^{M_j} z_{ij} n_{ijt} dj. \quad (4.4)$$

Letting $\tilde{N}_t \equiv \int_0^1 (\sum_i n_{ijt}) dj$ denote the overall amount of labor supplied, we can define aggregate TFP as

$$A_t \equiv \int_0^1 \left(\sum_{i=1}^{M_j} \frac{n_{ijt}}{\tilde{N}_t} z_{ij} \right) dj, \quad (4.5)$$

so that $Y_t \approx A_t \tilde{N}_t$. Equation (4.5) implies that a shift of employment from high- to low-productivity firms reduces aggregate TFP.²⁰ Since log changes in employment shares are proportional to log changes in wage-bill shares, a decline in the wage-bill share of high-productivity firms leads to a reduction in aggregate TFP.

Proposition 2 reveals the direction of the misallocation channel. In the presence of nominal rigidities, a rise in demand increases the wage-bill share of low-productivity firms and reduces that of high-productivity firms, leading to a decline in aggregate TFP. Two observations are worth emphasizing. First, misallocation arises only in the presence of heterogeneous firms. Firm heterogeneity is therefore not merely a modeling feature aimed at matching the micro evidence but a key element shaping the transmission of monetary policy under oligopsonistic competition. Second, our main finding is not that expansionary monetary policy reduces TFP, but that oligopsonistic competition dampens the effect of monetary policy through its effect on TFP. The overall response of TFP to monetary shocks may also depend on other mechanisms—such as changes in capital utilization—that are outside our framework and orthogonal to oligopsonistic competition.

The two channels we highlight—partial passthrough and misallocation—have been studied in some form in the context of imperfect competition in the product market. The partial-passthrough channel is present in [Mongey \(2021\)](#) and [Wang and Werning \(2022\)](#), who study monetary transmission under oligopolistic competition among homogeneous firms. The misallocation channel is present, in part, in [Baqae et al. \(2024\)](#), who analyze monetary policy when firms differ in both market power and price passthrough.²¹ One might therefore conjecture that imperfect competition has similar implications for monetary transmission regardless of whether it arises in product or labor markets. Our analysis shows that this is not the case.

Whether in the product market or in the labor market, imperfect competition generates partial and heterogeneous passthrough across firms. The difference in its implications for monetary policy stems from whether these passthroughs affect the supply or demand side of the market. Imperfect competition in the labor market—the focus of this paper—produces partial and heterogeneous passthrough in the demand for labor. Because households are on their labor-supply curve, partial passthrough

¹⁹The relationship holds only to a first order, because nominal price rigidities introduce a distortion ([Yun, 2005](#)).

²⁰The marginal product of labor is not equalized across firms in the non-stochastic steady state because the marginal disutility of labor is heterogeneous across firms—an implication of the nested-CES structure for the disutility of labor.

²¹In [Baqae et al. \(2024\)](#), firm heterogeneity is summarized by heterogeneity in markups and passthroughs, and misallocation reflects inefficient price dispersion across firms. In our setting, an analogous wage-dispersion channel is complemented by a direct reallocation of employment across firms with heterogeneous productivity.

to wages leads to a smaller employment response. In contrast, imperfect competition in the product market leads to partial passthrough of costs to prices, which results in a *larger*, not smaller, quantity response. Consequently, [Mongey \(2021\)](#) and [Wang and Werning \(2022\)](#) find that oligopolistic competition among homogeneous firms amplifies the output response to monetary policy. Similarly, [Baqaee et al. \(2024\)](#) find that lower passthrough by high-markup firms allows expansionary monetary policy to raise aggregate TFP by alleviating inefficient price dispersion.

Taken together, the stylized framework clarifies the core mechanisms through which oligopsonistic competition shapes the transmission of monetary policy. It highlights how nominal wage rigidities and convex markdowns jointly generate heterogeneous firm-level responses and dampen the passthrough of aggregate demand shocks. We next turn to the full model to quantify the strength of these channels and assess their implications for aggregate employment and output dynamics.

5 Quantitative Analysis

Oligopsonistic competition substantially dampens the real effects of monetary policy. In our calibrated model, output responds about 12 percent less than in a counterfactual economy where firms maintain constant markdowns, with partial passthrough accounting for roughly two-thirds of the attenuation and misallocation for the remaining third. This section develops the numerical method, calibrates the model, and unpacks the two channels in turn. We conclude by showing that the observed decline in local concentration since 1980 has amplified the real effects of monetary policy and reduced the inflation-activity trade-off faced by the central bank.

5.1 Numerical Solution

Most models of market power in the [Atkeson and Burstein \(2008\)](#) tradition are tractable because they involve a static game. Nominal rigidities introduce dynamics. A firm setting its wage has to consider future developments in both aggregate and local conditions, because it may not be able to respond to them when they are realized. This subsection discusses the challenges that arise due to the introduction of nominal rigidities and our approach to overcoming them.

Challenges. There are two main challenges for computing the equilibrium of the model. First, the law of large numbers does not apply to local labor markets with finitely many firms. Second, firms and workers interact in a large number of markets.

To appreciate the challenge posed by idiosyncratic uncertainty on local markets, consider the problem of finding the local equilibrium in a single market. Equation (3.20) shows that the optimal wage of firm i in market j at time t depends on expectations of aggregate variables $\{\mathbb{E}_{\mathcal{S}_{ijt}}[R_{t,t+\tau}], \mathbb{E}_{\mathcal{S}_{ijt}}[M_{t+\tau}]\}_{\tau=0}^{\infty}$ as well as the expectations of the local wage index $\{\mathbb{E}_{\mathcal{S}_{ijt}}[W_{jt+\tau}]\}_{\tau=0}^{\infty}$. Recall that the local wage index is given by

$$W_{jt+\tau} = \left(\sum_{i=1}^{M_j} W_{ijt+\tau}^{1+\eta} \right)^{\frac{1}{1+\eta}}, \quad W_{ijt+\tau} = \begin{cases} W_{ijt+\tau}^* & \text{if } \Theta_{ijt+\tau} = 1, \\ W_{ijt+\tau-1} & \text{if } \Theta_{ijt+\tau} = 0. \end{cases} \quad (5.1)$$

Since M_j is finite, the expected path of the local wage index until firm i can reoptimize its wage depends on the Calvo shocks of every other firm in market j . For concreteness, let us assume that there are 16 firms in the market, the median in our calibrated model, and assume that the probability that a wage spell lasts longer than 20 quarters is negligible. The result is $2^{300} \approx 2 \cdot 10^{90}$ possible permutations of wage adjustments. Evaluating this expectation exactly is impossible even for a single firm, not to mention finding the mutual best responses of 16 firms. And this is the problem for a single local market; the model has 10,000 of them.

The challenge of working with a large number of markets is that equilibrium prices are characterized implicitly, as solutions to a fixed-point problem. Having many markets means having to deal with a large-scale fixed-point problem. Our calibrated model features about 390,000 firms, inhabiting 10,000 markets. Solving for impulse responses over 100 quarters means characterizing $100 \times 390,000$ firm-level wages, $100 \times 10,000$ local wages plus 100×2 aggregate prices. This challenge is not present in typical HANK models, in which a distribution affects aggregate demand or supply but agents still interact on a small number of national markets. We have to both keep track of a high-dimensional distribution and solve for many equilibrium prices.

Solution. Since evaluating the expectations $\mathbb{E}_{\mathcal{S}_{ijt}}[W_{jt+\tau}]$ under uncertainty about Calvo shocks is infeasible, we endow firms with perfect foresight of both idiosyncratic as well as aggregate shocks.²² Firms learn the entire sequences of monetary policy shocks and Calvo shocks in period zero, and can compute the resulting paths of endogenous variables. All derivations in Section 3 and Appendix A.3 remain valid.

Our solution to the challenge of many markets exploits the fact that firms in market j do not compete directly with firms in market j' . This can be seen clearly from the labor supply curve

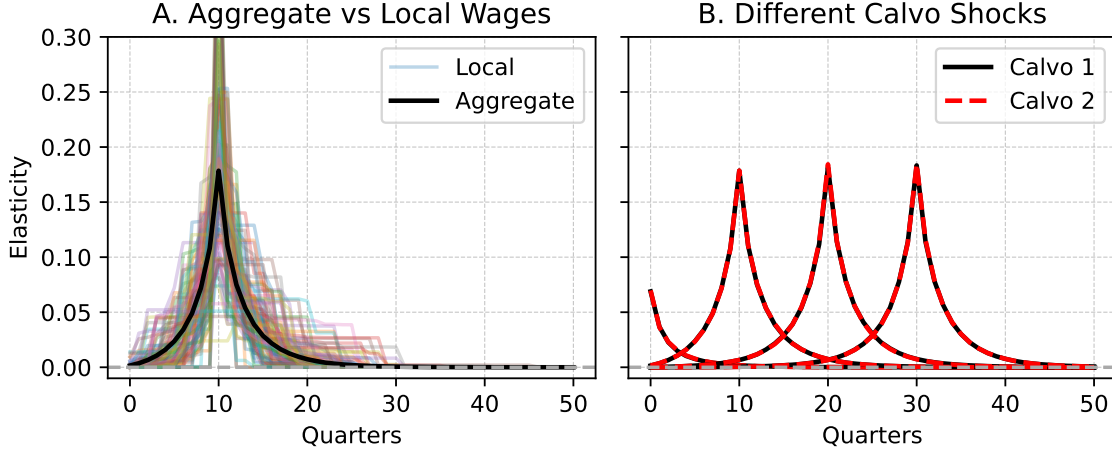
$$n_{ijt} = \left(\frac{W_{ijt}}{W_{jt}} \right)^\eta \left(\frac{W_{jt}}{W_t} \right)^\zeta N_t. \quad (5.2)$$

Firm ij considers how its wage W_{ijt} compares to the local wage W_{jt} and to the aggregate wage W_t ; local wages in other markets matter only through their effect on the aggregate wage. Given this insight, we solve the model in two steps. In the first step, we solve for Nash equilibria in local labor markets independently of each other and conditional on aggregates. The modularity of the SSJ method allows us to aggregate these local solutions into a single firm block. In the second step, we solve for general equilibrium in national markets.

Firm block. We use a large but finite number of markets, J , and assume that the economy returns to steady state after some finite number of periods T . The inner layer of the production block is a collection of J functions mapping $\{w_{jt}, \pi_t, m_t\}_{t=0}^T$ into $\{\tilde{w}_{jt}, A_{jt}\}_{t=0}^T$, where lower-case w 's denote real wages. The inputs are the payoff-relevant prices that firms take as given. The outputs are the local wages and productivity that result from firms' best response. Nash equilibrium corresponds

²²Assuming perfect foresight with respect to aggregate shocks is common in macroeconomics, especially in heterogeneous-agent models. Assuming perfect foresight with respect to idiosyncratic shocks is not common because it is typically not needed for tractability.

Figure 5: FIRM BLOCK JACOBIANS: WAGE WITH RESPECT TO SALES PRICE



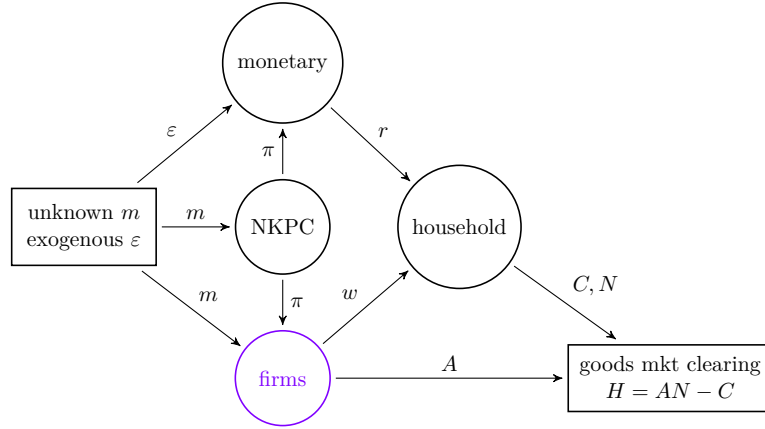
Notes: Panel A shows column 10 of the Jacobian of the local wage with respect to the sales price for a subsample of local labor markets (in color) as well as the corresponding response of the aggregate wage (in black). Panel B shows columns 0, 10, 20, and 30 of the Jacobian of the aggregate wage with respect to the sales price for two different sets of Calvo shocks.

to the fixed point $\tilde{w}_{jt} = w_{jt}$ for all t . In SSJ terminology, each local market is a “solved block”, $\{\pi_t, m_t\}_{t=0}^T \mapsto \{\tilde{w}_{jt}, A_{jt}\}_{t=0}^T$, whose Jacobians may be obtained by applying the implicit function theorem and the chain rule as usual. The last step is to aggregate the local solved Jacobians to obtain the Jacobians of the firm block as a whole, which maps aggregate prices $\{\pi_t, m_t\}_{t=0}^T$ into the aggregate wage and TFP $\{w_t, A_t\}_{t=0}^T$. In Appendix B, we walk through each of these steps in detail.

The key feature of our approach is to characterize the Jacobians of the firm block by manipulating many $T \times T$ matrices instead of a few $JT \times JT$ matrices, which would be infeasible for large J . We find that a large number of markets, $J = 10,000$, is required to ensure that impulse responses are not sensitive to further increasing the number of markets. Figure 5 illustrates the role of many markets. Panel A compares the Jacobians of local wages $\tilde{w}_{jt}$ for a random subsample of markets with the Jacobian of the aggregate wage w_t with respect to m_{10} , a rise in the sales price 10 quarters ahead. The responses of local wages are all different due to heterogeneity in the wage-bill shares and Calvo shocks of local firms. Furthermore, the local wage responses display visible steps that correspond to wage adjustments by a single large firm (or simultaneous adjustments by a few firms). In contrast, the Jacobian of the aggregate wage is smooth because aggregating over sufficiently many markets washes out the impact of individual Calvo shocks. Panel B makes this point more directly by showing that the Jacobians of the aggregate wage for two different sets of Calvo shocks are visually identical.

Full model. With the firm block in hand, we can represent the economy as a directed acyclic graph (DAG) shown by Figure 6. The DAG is a useful tool to conceptualize the monetary transmission mechanism as well as to solve the model numerically. Following the arrows reveals how the monetary policy shock ε propagates through the economy. The first-round effect of monetary easing—holding prices m and w fixed—is to lower the real interest rate r and stimulate consumption C . To meet the

Figure 6: DAG REPRESENTATION OF GENERAL EQUILIBRIUM



Notes: The diagram shows the dynamic acyclic graph representation of our New Keynesian model. The firm block is constructed by aggregating a large number of solved blocks, one for each local labor market. The Jacobians of the firm block capture the passthrough of shocks to the real sales price m_t and to inflation π_t into the aggregate real wage w_t and aggregate TFP A_t .

increase in aggregate demand, firms have to increase employment N by offering higher real wages w . The incentive to do so is provided by a rise in the relative price of intermediate goods m_t , which in turn raises inflation π and prompts the central bank to raise the nominal rate and curb the original stimulus. From a macroeconomic perspective, the oligopsonistic competition between heterogeneous firms simply modulates the increase in m required for aggregate supply to meet the increase in aggregate demand.

5.2 Calibration

We distinguish two sets of parameters. First, parameters that are common in New Keynesian models, which we set to conventional values. Second, parameters of the oligopsonistic firm block, which we calibrate internally to match moments of the distribution of firms and local labor markets. Table 2 summarizes the calibration. Table 3 shows the targeted moments that the model fits exactly.

Fixed parameters. We assume that the representative household has separable preferences over consumption and labor aggregates

$$U(C_t, N_t) = \frac{C_t^{1-\gamma}}{1-\gamma} - \varphi \frac{N_t^{1+\nu}}{1+\nu}, \quad (5.3)$$

and set the elasticity of intertemporal substitution to $1/\gamma = 0.5$, and the Frisch elasticity to $1/\nu = 0.5$. We normalize aggregate labor supply to $N = 1$ and back out the labor disutility parameter φ that justifies this as an equilibrium outcome. We set the quarterly discount factor to 0.99, implying an annual real interest rate of about 4 percent. Steady-state inflation is normalized to zero. The Taylor rule coefficient on inflation is $\phi_\pi = 1.5$ and we assume that the monetary policy shock follows an AR(1) process with quarterly autocorrelation $\rho_\epsilon = 0.85$. We set the elasticity of substitution between retail goods to $\epsilon = 7$, whose only role is to pin down the relative price of intermediate goods in steady state

Table 2: SUMMARY OF PARAMETERS

Parameter		Value	Parameter		Value
<i>A. Fixed parameters</i>			<i>B. Internally calibrated parameters</i>		
β	Discount factor	0.99	ζ	Across market elasticity	0.38
$1/\gamma$	EIS	0.50	η	Within market elasticity	2.50
$1/\nu$	Frisch elasticity	0.50	ξ_z	Productivity distribution shape	3.12
ϵ	Retail goods elasticity	7.00	ξ_m	Market size distribution shape	0.07
θ_p	Price stickiness	0.88			
θ_w	Wage stickiness	0.80			
ϕ_π	Taylor rule coefficient	1.50			
ρ_ϵ	Persistence of MP shock	0.85			

Notes: The table shows the parameters in our calibrated economy.

at $m = (\epsilon - 1)/\epsilon$. We set price stickiness to $\theta_p = 0.88$, within the range of typical estimates of price Phillips curves without indexation (see e.g., [Christiano et al. 2005](#); [Auclert et al. 2020](#)). Finally, we set wage stickiness to $\theta_w = 0.8$, consistent with the findings of [Grigsby et al. \(2021\)](#), who report that 80.6% of job stayers experience no wage change in a given quarter, and [Hazell and Taska \(forthcoming\)](#), who report that posted wages change every 5 quarters on average.

Internally calibrated parameters. The remaining parameters are the labor supply elasticities η and ζ as well as the parameters of the distributions of productivity z and market size M_j . We assume that the latter are independent Pareto distributions with shape parameters ξ_z and ξ_m , which leaves us with four parameters in total. We calibrate these four parameters by targeting an equal number of cross-sectional moments: the average markdown, the employment share of high-monopsony firms, the population share of high-monopsony firms, and mean HHI across local markets. Table 3 Panel A shows the values of these four moments in our baseline calibration, along with the parameters that they identify. These moments are endogenous outcomes of the model that, in principle, depend on all parameters. Nevertheless, there is a clear pattern of identification. Markdowns depend on the marginal revenue product of labor, which is not observable. Therefore, we rely on the estimates of [Yeh et al. \(2022\)](#) for U.S. manufacturing firms. They report an average markdown of 1.53, meaning that workers get paid 65 cents on the marginal dollar. The other three moments we compute directly from the LBD for the period 2015–2019. Less than 6 percent of firms are high monopsony—defined as having a wage-bill share above 10 percent—but they employ 40 percent of private sector workers. The average local labor market—defined as a county-subsector pair—is highly concentrated, with an HHI of 0.237. This value is close to what we would observe with four equal-sized firms. Panel B shows the moments we target in the counterfactual in Section 5.4.

Figure C.6 shows the steady-state distributions of market size and the wage-bill share implied by the calibration. The market size distribution is bimodal: duopsonies and effectively competitive markets with 100 firms are the two most common.²³ The wage-bill-share distribution is concentrated at the bottom—most firms have little market power—but has significant mass throughout the unit

²³We solve the model with 10,000 markets and truncate the number of firms per market at 100. Given the calibrated market size distribution, this yields more than 390,000 firms in total.

Table 3: TARGETED MOMENTS

Parameter	Corresponding Moment	Source	Value
<i>A. Baseline Calibration (2015–2019)</i>			
η	Average markdown	Yeh et al. (2022)	1.530
ζ	High-monopsony employment share	LBD	0.400
ξ_z	High-monopsony firm share	LBD	0.058
ξ_m	Mean local HHI	LBD	0.237
<i>B. Historical Calibration (1978–1980)</i>			
ξ_z	High-monopsony firm share	LBD	0.083
ξ_m	Mean local HHI	LBD	0.341

Notes: The table shows the targeted moments in our calibrated economy. The calibrated model matches each targeted moment exactly. Mean HHI is the average HHI across all local labor markets and years, calculated using firms’ wage-bill shares.

interval. Shares above 0.5 are possible only because of dispersion in firm-level productivity.

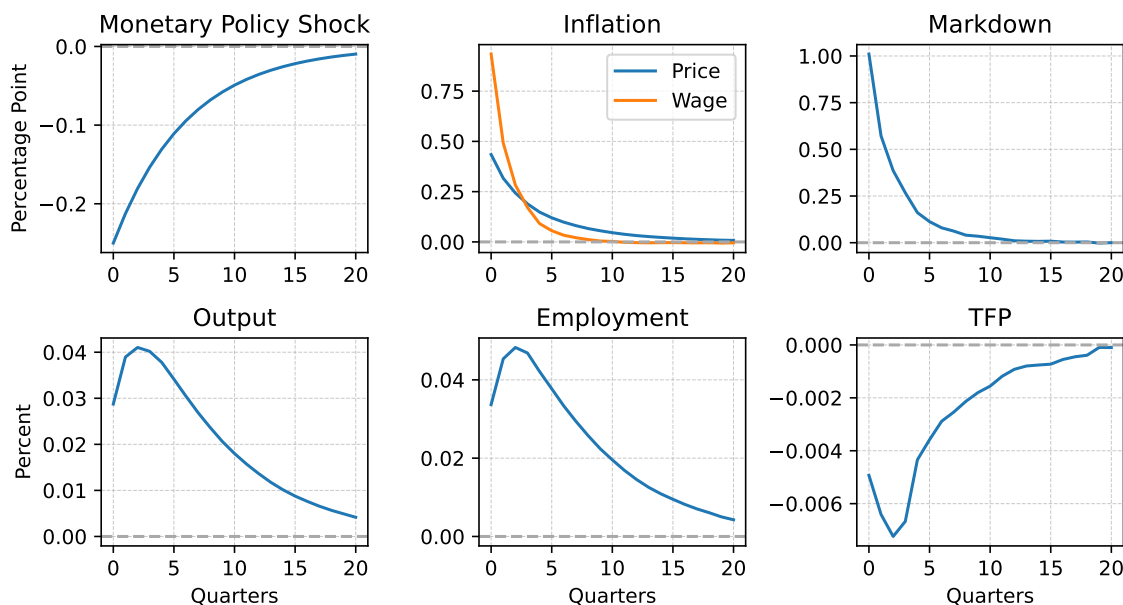
5.3 Labor Market Power and Monetary Policy

We now use the calibrated model to study how oligopsony shapes the transmission of monetary policy. We proceed in three steps: aggregate responses, heterogeneous responses across firms, and a decomposition into the partial-passthrough and misallocation channels.

Figure 7 shows the impulse responses of macroeconomic aggregates to a 25-basis point (annualized) expansionary monetary policy shock. The price responses in the top row are typical of standard New Keynesian models: price inflation rises by 40 basis points on impact and decays with the shock, and wage inflation rises about two times as much on impact but is much less persistent, reflecting a front-loaded rise in real wages. Despite the rise in real wages, the aggregate markdown $\mathcal{M}_t = 1 - w_t/(A_t m_t)$ increases substantially, implying that workers get paid less relative to the marginal revenue they generate. Inflation erodes the real wages paid by non-adjusting firms while the marginal revenue product of labor increases due to a rise in the intermediate good price; adjusters, in turn, raise their markdown optimally, because desired markdowns increase with market share. The bottom row of the figure shows that output and employment exhibit small hump shapes despite the absence of consumption habits. Aggregate employment rises by close to 0.05 percent at its peak, but output peaks at about 0.04 percent above its steady state. The final panel of the figure reveals the supply-side channel of monetary policy: Total factor productivity falls endogenously, dampening the response of output by 10 percent.

Figure 8 compares the model-implied heterogeneous response across firms with the empirical patterns in Section 2. Consistent with the mechanism highlighted in Section 4, the model reproduces our main empirical finding: after an expansionary monetary policy shock, low-monopsony firms expand their wage bills more than high-monopsony firms. Panel A plots the wage-bill responses of the two groups and shows that both respond positively, with the low-monopsony response about 50 percent

Figure 7: AGGREGATE RESPONSES TO AN EXPANSIONARY MONETARY POLICY SHOCK



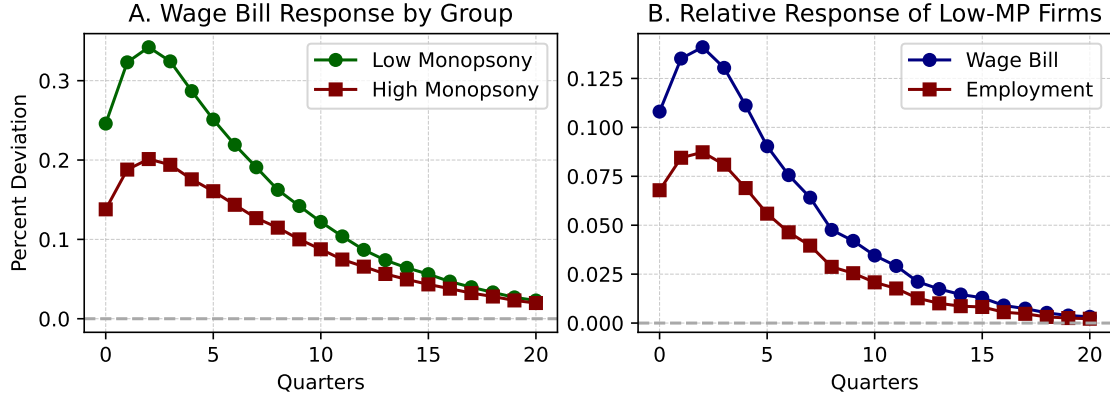
Notes: Figure shows aggregate responses to an expansionary monetary policy shock from our benchmark New Keynesian model with oligopsonistic wage setting. The first row contains the responses of rates, expressed in percentage point deviations from steady state. The second row contains the responses of quantities, expressed as percentage deviation from steady state.

larger in the first year after the shock. Panel B plots the relative wage-bill response together with the corresponding relative employment response, mirroring the empirical decomposition in Figure 3. These dynamic responses are not targeted in the calibration, which is disciplined only by steady-state moments of labor market structure. Quantitatively, the model understates the empirical differential: one year after the shock, the low-minus-high wage-bill response is 11 basis points in the model, below the 25–60 basis point peak range across empirical specifications. The model also generates faster dynamics, with wage bills peaking after about three quarters rather than two to three years in the data.²⁴

Partial passthrough and misallocation. Next, we consider a counterfactual that isolates the effects of oligopsonistic wage setting on monetary transmission. As Section 4 shows, oligopsony works through two channels, both of which operate through endogenous changes in firms’ desired markdown. Firms that can raise their wages gain market share at the expense of competitors whose wage is stuck. Adjusters’ desired markdown increases, and the passthrough becomes lower on average. This is the partial passthrough channel. Furthermore, the desired markdown of high-market-power firms falls more than the desired markdown of low-market-power firms. The allocation of employment shifts toward low-market power firms, which are less productive on average, generating a fall in aggregate

²⁴These gaps could be reduced by adding propagation and amplification forces that are absent from our baseline model. For example, consumption habits would slow the aggregate demand response and help generate more delayed wage-bill dynamics, while additional firm-level adjustment frictions or financial heterogeneity could amplify the cross-sectional differential. We leave these extensions for future work.

Figure 8: IMPULSE RESPONSES OF LOW- VS HIGH-MONOPSONY POWER FIRMS



Notes: Panel A shows the average wage bill responses of low- and high-monopsony power firms to a 25 basis points expansionary monetary policy shock, expressed as percentage deviation from the steady state. Panel B shows the wage bill and employment responses of low-monopsony-power firms relative to high-monopsony-power firms. As in the data, both groups of firms expand, but low-monopsony firms gain market share.

TFP. This is the misallocation channel. Therefore, we set up the counterfactual economy to be identical to the baseline in every respect, except that the desired markdowns of firms are constant. Formally, the wage-setting equation (3.20) becomes

$$W_{ijt}^* = \frac{\eta - (\eta - \zeta)\bar{s}_{ij}}{1 + \eta - (\eta - \zeta)\bar{s}_{ij}} \frac{\mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} n_{ijt+\tau} M_{t+\tau} z_{ij}}{\mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} n_{ijt+\tau}}, \quad (5.4)$$

where $\bar{s}_{ij}$ is the steady-state market share of firm ij .

The “constant desired markdown” model has the same steady state and admits the same DAG representation as our baseline model, which facilitates a clean analysis of the dynamic effects of monetary policy.²⁵ In both models, solving for local Nash equilibria conditional on national prices gives rise to a firm block of the form $\{m_t, \pi_t\}_{t=0}^{\infty} \mapsto \{w_t, A_t\}_{t=0}^{\infty}$. Figure C.7 compares the Jacobians of the benchmark and “constant desired markdown” versions of the firm block. As expected, the latter implies more responsive wages (no partial passthrough) and approximately zero TFP responses (no misallocation). Both channels of oligopsony are expected to dampen the effect of monetary policy on output. All else equal, partial passthrough implies that the sales price m_t has to rise more to induce firms to raise wages and employment to meet a given surge in aggregate demand, which leads to more inflation and prompts the central bank to lean against the boom. All else equal, misallocation implies that meeting a given surge in aggregate demand requires a larger increase in employment, which is only possible by paying higher wages to workers. Again, this requires a rise in the price of the intermediate good and generates inflation.

Table 4 shows how the real effects of monetary policy change as we turn on one or both channels of oligopsonistic wage setting. Specifically, the table reports the present value of the impulse response

²⁵Eliminating labor market power altogether would not only change the steady state but also require that wages be flexible; a profound change to the structure of the model that would complicate the analysis of the monetary transmission.

Table 4: OLIGOPSONY AND THE REAL EFFECTS OF MONETARY POLICY

	Output	Employment	TFP	Markdown
Constant Desired Markdown	1.00	1.00	0.00	5.31
Oligopsony	0.88	0.98	-0.10	5.77
Partial Passthrough	0.92	0.92	0.00	5.80
Misallocation	0.97	1.07	-0.10	5.30

Notes: The table reports the present value of the impulse responses of selected aggregate variables to an expansionary monetary policy shock. The responses are expressed in the same units as in Figure 7: percent deviation from steady state for output, employment, and TFP; percentage point deviation for the markdown. We use the output response in the constant-desired-markdown model as numeraire to make the results easier to interpret.

of output, employment, TFP, and the aggregate markdown, discounted with the steady-state real interest rate. The first row sets the reference point: in the model with fixed desired markdowns, an expansionary monetary policy shock increases output by 1 percent (a normalization), reflecting the same increase in employment and no change in TFP.²⁶ Even though the desired markdown of every firm is fixed by construction, the aggregate markdown rises by 5.31 percentage points. The real wages of non-adjusters are eroded by inflation while MRPL (which equals $z_{ij}m_t$) increases for all firms. Moreover, even adjusters do not fully pass through the initial rise in MRPL, given that they have to keep that nominal wage for 5 quarters, on average. In the full oligopsony model, output responds 12 percent less, resulting from a 2 basis point lower increase in employment and a 10 basis point decline in TFP. In addition, the aggregate markdown rises by 46 basis points more, reflecting the endogenous rise in desired markdowns among adjusters.

It is tempting to attribute the dampening entirely to misallocation. But in fact, both channels contribute, and partial passthrough accounts for the larger share. To isolate the role of partial passthrough, we solve a version of the model in which the Jacobians $\mathcal{J}^{w,m}$, $\mathcal{J}^{w,\pi}$ come from the oligopsony model but $\mathcal{J}^{A,m}$, $\mathcal{J}^{A,\pi}$ come from the constant-markdown model.²⁷ The third row shows that partial passthrough alone dampens the output response by 8 percent compared to the constant-markdown model. The fourth row shows that misallocation alone dampens the output response by 3 percent: the net effect of a 10% productivity loss and a 7% gain in employment. The misallocation channel alone dampens the rise of the average markdown, allowing real wages and employment to rise by more than in the constant-desired-markdown model. The intuition is that employment is reallocated towards low-market-power firms that are not only less productive but also have lower markdowns, which offset some of the TFP losses. Partial passthrough accounts for $(1 - 0.92)/(1 - 0.88) \approx 67\%$ of the combined effect, misallocation for $(1 - 0.97)/(1 - 0.88) \approx 25\%$, and the remaining 8% reflects the interaction between the two channels. The decomposition is not exact because the firm block's inputs $\{m_t, \pi_t\}$

²⁶Our chosen measures of the employment and TFP responses sum up to the output response because the aggregate production function $Y_t = A_t N_t$ is log-linear.

²⁷This corresponds to an economy in which wages respond to aggregate variables as in the benchmark economy, but TFP remains constant as in the constant-desired-markdown economy.

are endogenous.

5.4 Declining Local Concentration

The decline in U.S. local labor market concentration since 1980 has, through the lens of our model, made monetary policy roughly 10 percent more potent in stimulating output and accounts for about 15 percent of the estimated flattening of the Phillips curve over the same period. We establish these results by recalibrating the model to match concentration levels in 1978–1980 and comparing impulse responses and the Phillips multiplier—a measure of the inflation-activity trade-off proposed by [Barnichon and Mesters \(2021\)](#)—to the benchmark. Figure C.5 documents the trend in concentration using the two measures targeted by our model. The share of high-monopsony firms declined steadily from almost 9 percent in 1978 to less than 6 percent by 2015. The average HHI across local labor markets fell from 0.35 in 1978 to about 0.23 in 2000 and has remained roughly stable since.

The two calibrations differ only in the distribution of firms: we hold the primitive labor supply elasticities η and ζ fixed at their baseline values and recalibrate the Pareto shape parameters governing firm-level productivity and the number of firms per market to match the 1978–1980 high-monopsony firm share and mean local HHI in the LBD (Table 3, Panel B). We refer to this version as the *historical calibration*. Both channels of oligopsony are stronger in the historical calibration. It places more firms on the steeper part of the desired-markdown curve in Figure 4, which amplifies partial passthrough by raising the markdown adjustments of re-optimizing firms. It also features greater productivity dispersion across firms, which amplifies misallocation.

The decline in local concentration has raised the efficacy of monetary policy. Table 5 compares responses under the historical and benchmark calibrations. The benchmark output response is about 8 to 13 percent larger than in the historical calibration across horizons (column *Bench/Hist*), while the inflation response is about 3 percent smaller. The same monetary stimulus thus generates more output and less inflation, because both channels of oligopsony have weakened over time. Each channel raises the price increase required to elicit a given expansion: partial passthrough leads firms to absorb part of a price rise as higher markdowns rather than passing it through to wages, while misallocation forces employment to rise more than output to overcome the endogenous decline in TFP, which itself requires higher wages. A larger required price increase means more inflation per unit of output—and both channels are weaker in the benchmark than in the historical calibration.

It follows that the decline in concentration has reduced the inflation-activity trade-off. We measure the trade-off using the (expectations-augmented) Phillips multiplier of [Barnichon and Mesters \(2021\)](#), the change in average inflation, net of expected inflation, per unit change in average output induced by a monetary policy shock:

$$\mathcal{P}_h = \frac{d(\bar{\pi}_{t:t+h} - \bar{\pi}_{t:t+h}^e)}{d\varepsilon_{t:\infty}} \bigg/ \frac{d\bar{y}_{t:t+h}}{d\varepsilon_{t:\infty}}, \quad (5.5)$$

where $\bar{x}_{t:t+h} \equiv \frac{1}{h+1} \sum_{k=0}^h x_{t+k}$ and $\pi_t^e \equiv \mathbb{E}_t[\pi_{t+1}]$ is the one-period-ahead inflation expectation. This multiplier is a generalization of the slope of the Phillips curve: in a textbook New Keynesian model with $\pi_t = \kappa y_t + \mathbb{E}_t[\pi_{t+1}]$, $\mathcal{P}_h = \kappa$ for all h . In our model, sticky prices, sticky wages, and heterogeneous firms enrich the dynamics of the inflation-output relationship, so $\mathcal{P}_h$ varies with h , but it remains a

Table 5: DECLINING CONCENTRATION AND THE EFFICACY OF MONETARY POLICY

		Benchmark	Historical	Bench/Hist
Output response (%)	$h = 4$	0.04	0.03	1.13
	$h = 8$	0.04	0.03	1.10
	$h = 20$	0.02	0.02	1.08
Inflation response (pp)	$h = 4$	0.07	0.08	0.97
	$h = 8$	0.05	0.05	0.97
	$h = 20$	0.03	0.03	0.97
Phillips multiplier ($\mathcal{P}_h$)	$h = 4$	0.48	0.55	0.87
	$h = 8$	0.33	0.37	0.90
	$h = 20$	0.25	0.28	0.91

Notes: The table compares the benchmark calibration, which matches U.S. local labor market concentration in 2015–2019, with the historical calibration, which matches 1978–1980 (Table 3, Panel B). The blocks report responses to a 25 basis point (annualized) expansionary monetary policy shock at horizons h , measured in quarters: the output response (percent deviation from steady state), the inflation response (percentage point deviation), and the Phillips multiplier $\mathcal{P}_h$, defined in the text. The final column reports the ratio of the benchmark to the historical value.

counterpart to the Phillips-curve slope.

Table 5 reports $\mathcal{P}_h$ under both calibrations. The model-implied slope sits squarely within the range of empirical estimates. For example, Blanchard (2016) estimates that the slope of the unemployment-gap-based Phillips curve fell from about 0.8 to 0.3 between 1980 and 2013, which Okun’s law translates to a fall from 0.4 to 0.15 in output-gap terms. Our model’s Phillips multiplier runs from about 0.37 to 0.25 at medium to long horizons, and from 0.55 to 0.48 at the one-year horizon. Furthermore, the decline in concentration moves the slope in the right direction, accounting for about 15 percent of the estimated flattening between 1980 and the 2010s—a non-negligible contribution from a single, newly identified channel that operates alongside other forces, such as better-anchored inflation expectations. Whether a flatter Phillips curve is welcome depends on the policy context: in an easing episode, a smaller trade-off lets the central bank deliver more output stimulus for a given amount of inflation; in a tightening episode, achieving a given reduction in inflation requires a larger output sacrifice.

6 Conclusion

This paper studies how local labor market concentration affects the transmission of monetary policy to macroeconomic aggregates. Using administrative U.S. Census data, we provide evidence that firms respond differently to an identified monetary policy shock according to their wage-bill share in the local labor market. Specifically, we estimate that the wage bill of firms with a market share below 10 percent responds about 40 percent more than that of firms above that cutoff. We propose a New Keynesian model with oligopsonistic competition which can account for the heterogeneous response across firms. To deliver this result, our model requires (i) nominal wage stickiness and (ii) that the

superelasticity of labor supply is decreasing in the wage-bill share. If both conditions are met, then monetary easing raises the desired markdown of firms, dampening the passthrough of prices to wages. If high-monopsony-power firms are more productive on average, then aggregate TFP falls as well. According to our calibrated model, oligopsonistic wage setting reduces the impact of monetary policy on output by 12 percent, with two-thirds of the difference attributed to partial passthrough and about one-third to misallocation. Finally, our model also implies that the observed decline in local labor market concentration in the U.S. since 1980 has amplified the effect of monetary policy on output and contributed to the flattening of the Phillips curve.

In our model, monetary policy affects firms indirectly by stimulating aggregate demand. In future work, it would be interesting to extend our analysis by allowing monetary policy to affect firms' financing directly. Furthermore, some of the aggregate implications of labor market power, notably weaker real effects of monetary policy and countercyclical TFP, go against the implications of pure product market oligopoly. Studying labor and product market power in a unified framework is another exciting avenue for future research.

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Online Appendix

A Mathematical Derivations and Proofs

A.1 Household's problem

The representative household solves

$$\max_{\{c_{kt}, n_{ijt}, B_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t) \quad (\text{A.1})$$

$$\text{s.t.} \quad \int_0^1 p_{kt} c_{kt} dk + B_t = (1 + i_{t-1})B_{t-1} + \int_0^1 \sum_{i=1}^{M_j} W_{ijt} n_{ijt} dj + \Pi_t, \quad (\text{A.2})$$

$$C_t = \left(\int_0^1 c_{kt}^{\frac{\epsilon-1}{\epsilon}} dk \right)^{\frac{\epsilon}{\epsilon-1}}, \quad N_t = \left(\int_0^1 n_{jt}^{\frac{\zeta+1}{\zeta}} dj \right)^{\frac{\zeta}{\zeta+1}}, \quad n_{jt} = \left(\sum_{i=1}^{M_j} n_{ijt}^{\frac{\eta+1}{\eta}} \right)^{\frac{\eta}{\eta+1}}. \quad (\text{A.3})$$

We solve this problem in four steps.

Step 1. Derive retailer-specific demand curves and aggregate price index from a static expenditure minimization problem.

$$P_t C_t = \min_{\{c_{kt}\}} \int_0^1 p_{kt} c_{kt} dk \quad \text{s.t.} \quad \left(\int_0^1 c_{kt}^{\frac{\epsilon-1}{\epsilon}} dk \right)^{\frac{\epsilon}{\epsilon-1}} = \bar{U} \quad (\text{A.4})$$

Let λ_t denote the Lagrange multiplier on the constraint. The FOC is

$$p_{kt} = \lambda_t C_t^{\frac{1}{\epsilon}} c_{kt}^{-\frac{1}{\epsilon}} \quad (\text{A.5})$$

Integrating across varieties yields

$$\int_0^1 p_{kt} c_{kt} dk = \lambda_t C_t^{\frac{1}{\epsilon}} \int_0^1 c_{kt}^{\frac{\epsilon-1}{\epsilon}} dk = \lambda_t C_t \quad (\text{A.6})$$

Combining this last expression with the definition of the price index as $P_t C_t = \int_0^1 p_{kt} c_{kt} dk$ implies that $P_t = \lambda_t$. Substitute this into the FOC to get the retailer-specific demand curve

$$c_{kt} = \left(\frac{p_{kt}}{P_t} \right)^{-\epsilon} C_t \quad (\text{A.7})$$

Substitute the demand curve back into the FOC to express the price index

$$p_{kt}c_{kt} = p_{kt}^{1-\epsilon} P_t^\epsilon C_t \quad (\text{A.8})$$

$$\int_0^1 p_{kt}c_{kt} dk = P_t^\epsilon C_t \int_0^1 p_{kt}^{1-\epsilon} dk \quad (\text{A.9})$$

$$P_t C_t = P_t^\epsilon C_t \int_0^1 p_{kt}^{1-\epsilon} dk \quad (\text{A.10})$$

$$P_t = \left(\int_0^1 p_{kt}^{1-\epsilon} dk \right)^{\frac{1}{1-\epsilon}} \quad (\text{A.11})$$

Step 2. Derive local labor-market specific labor supply and aggregate wage index from static income maximization problem.

$$W_t N_t = \max_{\{n_{jt}\}} \int_0^1 W_{jt} n_{jt} dj \quad \text{s.t.} \quad \left(\int_0^1 n_{jt}^{\frac{\zeta+1}{\zeta}} dj \right)^{\frac{\zeta}{\zeta+1}} = N_t \quad (\text{A.12})$$

Let λ_t denote the Lagrange multiplier on the constraint. The FOC is

$$W_{jt} = \lambda_t N_t^{-\frac{1}{\zeta}} n_{jt}^{\frac{1}{\zeta}} \quad (\text{A.13})$$

Integrating across local labor markets yields

$$\int_0^1 W_{jt} n_{jt} dj = \lambda_t N_t^{-\frac{1}{\zeta}} \int_0^1 n_{jt}^{\frac{\zeta+1}{\zeta}} dj = \lambda_t N_t \quad (\text{A.14})$$

Combining this last expression with the definition of the aggregate wage index as $W_t N_t = \int_0^1 W_{jt} n_{jt} dj$ implies that $W_t = \lambda_t$. Substitute this into the FOC to get the local labor market-specific labor supply curve

$$n_{jt} = \left(\frac{W_{jt}}{W_t} \right)^\zeta N_t \quad (\text{A.15})$$

Substitute the labor supply curve back into the FOC to express the aggregate wage index

$$W_{jt} n_{jt} = W_{jt}^{1+\zeta} W_t^{-\zeta} N_t \quad (\text{A.16})$$

$$\int_0^1 W_{jt} n_{jt} dj = W_t^{-\zeta} N_t \int_0^1 W_{jt}^{1+\zeta} dj \quad (\text{A.17})$$

$$W_t N_t = W_t^{-\zeta} N_t \int_0^1 W_{jt}^{1+\zeta} dj \quad (\text{A.18})$$

$$W_t = \left(\int_0^1 W_{jt}^{1+\zeta} dj \right)^{\frac{1}{1+\zeta}} \quad (\text{A.19})$$

Step 3. Derive firm-specific labor supply and local wage indexes from static income maximization problem.

$$W_{jt}n_{jt} = \max_{\{n_{ijt}\}} \sum_{i=1}^{M_j} W_{ijt}n_{ijt} \quad \text{s.t.} \quad \left(\sum_{i=1}^M n_{ijt}^{\frac{\eta+1}{\eta}} \right)^{\frac{\eta}{\eta+1}} = n_{jt} \quad (\text{A.20})$$

Analogously to the second step, we obtain

$$n_{ijt} = \left(\frac{W_{ijt}}{W_{jt}} \right)^{\eta} n_{jt}, \quad W_{jt} = \left(\sum_{i=1}^{M_j} W_{ijt}^{1+\eta} \right)^{\frac{1}{1+\eta}}. \quad (\text{A.21})$$

Combining (A.15) and (A.21) implies the firm-specific labor supply curve in the main text

$$n_{ijt} = \left(\frac{W_{ijt}}{W_{jt}} \right)^{\eta} \left(\frac{W_{jt}}{W_t} \right)^{\zeta} N_t \quad (\text{A.22})$$

Step 4. Derive aggregate consumption, labor supply, and savings choices from dynamic utility maximization problem.

$$\max_{C_t, N_t, B_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t) \quad \text{s.t.} \quad P_t C_t + B_t = (1 + i_{t-1})B_{t-1} + W_t N_t + \Pi_t \quad (\text{A.23})$$

Let λ_t denote the Lagrange multipliers on the budget constraint for all $t \geq 0$. The FOCs are

$$[C_t]: \quad 0 = U_C(C_t, N_t) - \lambda_t P_t \quad (\text{A.24})$$

$$[N_t]: \quad 0 = U_N(C_t, N_t) + \lambda_t W_t \quad (\text{A.25})$$

$$[B_t]: \quad 0 = -\lambda_t + \beta \mathbb{E}_t[(1 + i_t)\lambda_{t+1}] \quad (\text{A.26})$$

Combining the FOCs yields the aggregate Euler equation and labor supply curves

$$U_C(C_t, N_t) = \beta \mathbb{E}_t \left[\frac{1 + i_t}{\pi_{t+1}} U_C(C_{t+1}, N_{t+1}) \right], \quad \frac{W_t}{P_t} = -\frac{U_N(C_t, N_t)}{U_C(C_t, N_t)} \quad (\text{A.27})$$

A.2 Retailers

Firm k solves

$$\max_{p_{kt}^*, \{c_{kt+\tau}, y_{kt+\tau}\}} \sum_{\tau=0}^{\infty} \theta_p^{\tau} R_{t,t+\tau} (p_{kt}^* c_{kt+\tau} - M_{t+\tau} y_{kt+\tau}) \quad (\text{A.28})$$

$$\text{s.t.} \quad c_{kt+\tau} = y_{kt+\tau} \quad (\text{A.29})$$

$$c_{kt+\tau} = \left(\frac{p_{kt}^*}{P_{t+\tau}} \right)^{-\epsilon} C_{t+\tau} \quad (\text{A.30})$$

Substitute the constraints into the objective function

$$\sum_{\tau=0}^{\infty} R_{t,t+\tau} C_{t+\tau} P_{t+\tau}^{\epsilon} \left[(p_{kt}^*)^{1-\epsilon} - M_{t+\tau} (p_{kt}^*)^{-\epsilon} \right]$$

The FOC with respect to p_{kt}^* is

$$0 = \sum_{\tau=0}^{\infty} \theta_p^{\tau} R_{t,t+\tau} C_{t+\tau} P_{t+\tau}^{\epsilon} \left[(1-\epsilon)(p_{kt}^*)^{-\epsilon} + \epsilon M_{t+\tau} (p_{kt}^*)^{-\epsilon-1} \right] \quad (\text{A.31})$$

$$0 = \sum_{\tau=0}^{\infty} \theta_p^{\tau} R_{t,t+\tau} C_{t+\tau} P_{t+\tau}^{\epsilon} [(1-\epsilon)p_{kt}^* + \epsilon M_{t+\tau}] \quad (\text{A.32})$$

$$p_{kt}^* = \frac{\epsilon}{\epsilon-1} \frac{\sum_{\tau=0}^{\infty} \theta_p^{\tau} R_{t,t+\tau} P_{t+\tau}^{\epsilon} C_{t+\tau} M_{t+\tau}}{\sum_{\tau=0}^{\infty} \theta_p^{\tau} R_{t,t+\tau} P_{t+\tau}^{\epsilon} C_{t+\tau}} \quad (\text{A.33})$$

Given that Calvo shocks are iid and the price index is given by

$$P_t = \left(\int_0^1 p_{kt}^{1-\epsilon} dk \right)^{\frac{1}{1-\epsilon}} \quad (\text{A.34})$$

we can write the law of motion for the price level as

$$P_t = \left[(1-\theta_p)(p_{kt}^*)^{1-\epsilon} + \theta_p P_{t-1}^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \quad (\text{A.35})$$

Phillips curve. Divide (A.33) and (A.35) with P_t to eliminate nominal variables

$$\frac{p_{kt}^*}{P_t} = \frac{\epsilon}{\epsilon-1} \frac{\sum_{\tau=0}^{\infty} \theta_p^{\tau} R_{t,t+\tau} \pi_{t,t+\tau}^{\epsilon+1} C_{t+\tau} m_{t+\tau}}{\sum_{\tau=0}^{\infty} \theta_p^{\tau} R_{t,t+\tau} \pi_{t,t+\tau}^{\epsilon} C_{t+\tau}} \quad (\text{A.36})$$

$$\frac{p_t^*}{P_t} = \left(\frac{1 - \theta_p \pi_{t-1,t}^{\epsilon-1}}{1 - \theta_p} \right)^{\frac{1}{1-\epsilon}} \quad (\text{A.37})$$

where $m_t \equiv M_t/P_t$ denotes real marginal cost. We can express inflation recursively as

$$\frac{K_t}{F_t} = \left(\frac{1 - \theta_p \pi_{t-1,t}^{\epsilon-1}}{1 - \theta_p} \right)^{\frac{1}{1-\epsilon}} \quad (\text{A.38})$$

$$K_t = \epsilon C_t m_t + \theta_p \frac{\pi_{t,t+1}^{\epsilon+1}}{1 + i_t} K_{t+1} \quad (\text{A.39})$$

$$F_t = (\epsilon-1)C_t + \theta_p \frac{\pi_{t,t+1}^{\epsilon}}{1 + i_t} F_{t+1} \quad (\text{A.40})$$

Loglinearize these equations, assuming $\pi_{ss} = 1$

$$\hat{K}_t - \hat{F}_t = \frac{\theta_p}{1 - \theta_p} \hat{\pi}_{t-1,t} \quad (\text{A.41})$$

$$\hat{K}_t = (1 - \beta\theta_p) (\hat{C}_t + \hat{m}_t) + \beta\theta_p [(\epsilon + 1)\hat{\pi}_{t,t+1} + \hat{K}_{t+1} - \beta d i_t] \quad (\text{A.42})$$

$$\hat{F}_t = (1 - \beta\theta_p) \hat{C}_t + \beta\theta_p [\epsilon \hat{\pi}_{t,t+1} + \hat{F}_{t+1} - \beta d i_t] \quad (\text{A.43})$$

Combine these to get the textbook NKPC

$$\frac{\theta_p}{1 - \theta_p} \hat{\pi}_{t-1,t} = (1 - \beta\theta_p) \hat{m}_t + \beta\theta_p \left[\hat{\pi}_{t,t+1} + \frac{\theta_p}{1 - \theta_p} \hat{\pi}_{t,t+1} \right] \quad (\text{A.44})$$

$$\hat{\pi}_{t-1,t} = \frac{(1 - \theta_p)(1 - \beta\theta_p)}{\theta_p} \hat{m}_t + \beta \hat{\pi}_{t,t+1} \quad (\text{A.45})$$

A.3 Producers

We start from the decision problem of a generic producer ij that contemplates its wage strategy for every possible state of the world. Then, we show that the wage setting problem is separable across markets and wage spells, which greatly simplifies the optimization problem. Finally, we derive the wage setting equation.

Full problem. The state of the world s_t in period t consists of the monetary policy shock ϵ_t as well as the Calvo shocks for all firms $\{\{\Theta_{ijt}\}_i\}_j$, where Θ_{ijt} equals one when firm i in sector j can adjust its wage at time t , and zero otherwise. We denote by s^t the history of states, $s^t = \{s_0, \dots, s_t\}$. The problem of an oligopolistic intermediate good producer is

$$\max_{\{W_{ij}(s^t), n_{ij}(s^t), W_j(s^t)\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \sum_{s^t} R(s^t) \left[M(s^t) z_{ij} n_{ij}(s^t) - n_{ij}(s^t) W_{ij}(s^t) \right], \quad (\text{A.46})$$

$$\text{s.t.} \quad n_{ij}(s^t) = \left(\frac{W_{ij}(s^t)}{W_j(s^t)} \right)^\eta \left(\frac{W_j(s^t)}{W(s^t)} \right)^\zeta N(s^t), \quad (\text{A.47})$$

$$W_j(s^t) = \left[W_{ij}(s^t)^{1+\eta} + \sum_{i' \neq i} W_{i'j}(s^t)^{1+\eta} \right]^{\frac{1}{1+\eta}}, \quad (\text{A.48})$$

$$W_{ij}(s^t) = W_{ij}(s^{t-1}) \quad \text{if } \Theta_{ijt} = 0. \quad (\text{A.49})$$

Note that the firm takes as given $\{R(s^t), M(s^t), W(s^t), N(s^t), W_{i'j}(s^t)\}$ when choosing their actions.

Separability across markets. Lemma 1 shows that a firm's problem depends only on the history of the aggregate state and the Calvo states on their own market. That is, Calvo realizations affect local outcomes, but local markets interact with each other only via aggregate prices and quantities.

LEMMA 1. *Let $s_\epsilon^t = \{\epsilon_0, \dots, \epsilon_t\}$ denote the history of the aggregate state, and $s_j^t = \{\{\epsilon_0, \{\Theta_{ij0}\}_i\}, \dots, \{\epsilon_t, \{\Theta_{ijt}\}_i\}\}$ denote the history of the aggregate state and the Calvo states for the specific market j*

where the firm operates. The firm's problem can be rewritten as

$$\begin{aligned}
& \max_{\{W_{ij}(s_j^t), n_{ij}(s_j^t), W_j(s_j^t)\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \sum_{s_j^t} R(s_\epsilon^t) \left[M(s_\epsilon^t) z_{ij} n_{ij}(s_j^t) - n_{ij}(s_j^t) W_{ij}(s_j^t) \right], \\
& \text{s.t.} \quad n_{ij}(s_j^t) = \left(\frac{W_{ij}(s_j^t)}{W_j(s_j^t)} \right)^\eta \left(\frac{W_j(s_j^t)}{W(s_\epsilon^t)} \right)^\zeta N(s_\epsilon^t), \\
& \quad W_j(s_j^t) = \left[W_{ij}(s_j^t)^{1+\eta} + \sum_{i' \neq i} W_{i'j}(s_j^t)^{1+\eta} \right]^{\frac{1}{1+\eta}}, \\
& \quad W_{ij}(s_j^t) = W_{ij}(s_j^{t-1}) \quad \text{if } \Theta_{ijt} = 0.
\end{aligned} \tag{A.50}$$

Proof. Conditional on the aggregate shock history s_ϵ^t , the continuum of local markets implies that idiosyncratic Calvo realizations outside market j wash out in the aggregate. Hence aggregate objects such as $R(s_\epsilon^t)$, $M(s_\epsilon^t)$, $W(s_\epsilon^t)$, and $N(s_\epsilon^t)$ are functions only of the aggregate history. Firm ij takes these aggregate objects and the wage strategies of firms outside market j as given. Its payoff-relevant local state is therefore only the history of Calvo realizations within its own market, together with the aggregate history. Thus the firm's payoff and constraints can be written as functions of s_j^t and s_ϵ^t . ■

Separability across wage spells. Next, we argue that a firm that is allowed to adjust its wage need only to consider the states of the world until its next wage adjustment.

Given Definition 1, we can partition the firm's problem into all different wage spells:

$$\max_{\{W_{ij}(\mathcal{S}_{ijt}), n_{ij}(s_j^{t+\tau}), W_j(s_j^{t+\tau})\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \sum_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} \sum_{s^{t+\tau} \in \mathcal{S}_{ijt}} R(s_\epsilon^{t+\tau}) \left[M(s_\epsilon^{t+\tau}) z_{ij} n_{ij}(s_j^{t+\tau}) - n_{ij}(s_j^{t+\tau}) W_{ij}(\mathcal{S}_{ijt}) \right], \tag{A.51}$$

$$\text{s.t.} \quad n_{ij}(s_j^{t+\tau}) = \left(\frac{W_{ij}(\mathcal{S}_{ijt})}{W_j(s_j^{t+\tau})} \right)^\eta \left(\frac{W_j(s_j^{t+\tau})}{W(s_\epsilon^{t+\tau})} \right)^\zeta N(s_\epsilon^{t+\tau}), \tag{A.52}$$

$$W_j(s_j^{t+\tau}) = \left[W_{ij}(\mathcal{S}_{ijt})^{1+\eta} + \sum_{i' \neq i} W_{i'j}(s_j^{t+\tau})^{1+\eta} \right]^{\frac{1}{1+\eta}}, \tag{A.53}$$

where we have used that a firm's own wage cannot vary within a wage spell but its employment might be due to aggregate shocks and changes in the wages of its competitors. We make two observations. First, the firm takes as given all payoff-relevant prices and quantities except their own wage. Second, the firm's own wage is a state variable within wage spells, but has no effect on past or future wage spells.

It follows that we may focus on setting an optimal wage W^* for a single generic wage spell $\mathcal{S}_{ijt}$.

$$\max_{W_{ijt}^*, \{n_{ij}(s^{t+\tau}), W_j(s^{t+\tau})\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \sum_{s^{t+\tau} \in \mathcal{S}_{ijt}} R(s^{t+\tau}) \left[M(s^{t+\tau}) z_{ij} n_{ij}(s^{t+\tau}) - n_{ij}(s^{t+\tau}) W_{ijt}^* \right], \quad (\text{A.54})$$

$$\text{s.t.} \quad n_{ij}(s^{t+\tau}) = \left(\frac{W_{ijt}^*}{W_j(s^{t+\tau})} \right)^\eta \left(\frac{W_j(s^{t+\tau})}{W(s^{t+\tau})} \right)^\zeta N(s^{t+\tau}), \quad (\text{A.55})$$

$$W_j(s^{t+\tau}) = \left[\left(W_{ijt}^* \right)^{1+\eta} + \sum_{i' \neq i} W_{i'j}(s^{t+\tau})^{1+\eta} \right]^{\frac{1}{1+\eta}}. \quad (\text{A.56})$$

To emphasize the similarity to the standard Calvo price setting problem of the retailers, we fold the dependence on the histories of states into the expectation operator. The wage spell-specific expectation of a generic variable X is defined as

$$\mathbb{E}_{\mathcal{S}_{ijt}}[X_{t+\tau}] \equiv \sum_{s^{t+\tau} \in \mathcal{S}_{ijt}} P(s^{t+\tau} | s^t) X(s^{t+\tau}), \quad (\text{A.57})$$

where $P(s^{t+\tau} | s^t)$ is the probability of reaching history $s^{t+\tau}$ from history s^t . This operator allows us to write the wage setting problem more compactly as

$$\max_{\{W_{ijt}^*, n_{ijt+\tau}, W_{jt+\tau}\}} \mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} \left[M_{t+\tau} z_{ij} n_{ijt+\tau} - n_{ijt+\tau} W_{ijt}^* \right], \quad (\text{A.58})$$

$$\text{s.t.} \quad n_{ijt+\tau} = \left(\frac{W_{ijt}^*}{W_{jt+\tau}} \right)^\eta \left(\frac{W_{jt+\tau}}{W_{t+\tau}} \right)^\zeta N_{t+\tau}, \quad (\text{A.59})$$

$$W_{jt+\tau} = \left[\left(W_{ijt}^* \right)^{1+\eta} + \sum_{i' \neq i} W_{i'j}(s^{t+\tau})^{1+\eta} \right]^{\frac{1}{1+\eta}}. \quad (\text{A.60})$$

First order conditions. Let $\lambda_{ijt+\tau}$ denote the Lagrange multiplier on labor supply, and $\nu_{ijt+\tau}$ denote the Lagrange multiplier on the wage index. The FOCs are

$$[W_{ijt}^*]: \quad \mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} W_{ijt}^* n_{ijt+\tau} = \mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} \left[\eta n_{ijt+\tau} \lambda_{ijt+\tau} - W_{jt+\tau}^{-\eta} \left(W_{ijt}^* \right)^{1+\eta} \nu_{ijt+\tau} \right] \quad (\text{A.61})$$

$$[W_{jt+\tau}]: \quad (\eta - \zeta) n_{ijt+\tau} \lambda_{ijt+\tau} = \nu_{ijt+\tau} W_{jt+\tau} \quad (\text{A.62})$$

$$[n_{ijt+\tau}]: \quad R_{t,t+\tau} \left(M_{t+\tau} z_{ij} - W_{ijt}^* \right) = \lambda_{ijt+\tau} \quad (\text{A.63})$$

Combine all three

$$\begin{aligned}
\mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} W_{ijt}^* n_{ijt+\tau} &= \mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} \left[\eta n_{ijt+\tau} \lambda_{ijt+\tau} - \left(\frac{W_{ijt}^*}{W_{jt+\tau}} \right)^{1+\eta} \nu_{ijt+\tau} W_{jt+\tau} \right] \\
&= \mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} n_{ijt+\tau} \lambda_{ijt+\tau} \left[\eta - (\eta - \zeta) \left(\frac{W_{ijt}^*}{W_{jt+\tau}} \right)^{1+\eta} \right] \\
&= \mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} \left(M_{t+\tau} z_{ij} n_{ijt+\tau} - W_{ijt}^* n_{ijt+\tau} \right) \left[\eta - (\eta - \zeta) \left(\frac{W_{ijt}^*}{W_{jt+\tau}} \right)^{1+\eta} \right]
\end{aligned}$$

which we can rearrange to get

$$W_{ijt}^* = \frac{\mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} n_{ijt+\tau} \left[\eta - (\eta - \zeta) \left(\frac{W_{ijt}^*}{W_{jt+\tau}} \right)^{1+\eta} \right] M_{t+\tau} z_{ij}}{\mathbb{E}_{\mathcal{S}_{ijt}} \sum_{\tau=0}^{\infty} R_{t,t+\tau} n_{ijt+\tau} \left[1 + \eta - (\eta - \zeta) \left(\frac{W_{ijt}^*}{W_{jt+\tau}} \right)^{1+\eta} \right]} \quad (\text{A.64})$$

Wage-bill share. Next, we prove that $\left(\frac{W_{ijt}}{W_{jt}} \right)^{1+\eta}$ is the wage-bill share of firm ij in period t .

$$s_{ijt} \equiv \frac{W_{ijt} n_{ijt}}{\sum_{l=1}^{M_j} W_{ljt} n_{ljt}} = \frac{W_{ijt} n_{ijt}}{\sum_{l=1}^{M_j} W_{ljt} n_{ijt} \left(\frac{W_{ijt}}{W_{ljt}} \right)^{-\eta}} = \frac{W_{ijt}^{1+\eta}}{\sum_l W_{ljt}^{1+\eta}} = \frac{W_{ijt}^{1+\eta}}{W_{jt}^{1+\eta}} \quad (\text{A.65})$$

The first equality is the definition of the local wage-bill share. The second equality uses the ratio of labor supply curves to firms i and l . The third equality just collects and cancels terms. The fourth equality uses the expression of the local wage index.

A.4 Special case

This section contains the proofs of Propositions 1 and 2.

Proof of Proposition 1. Proof. Conjecture that there exists a constant factor $A > 0$ such that $W_{i,1}^* = A W_{i,0}^*$ for all i . The conjecture implies that the local wage index increases by the same factor, $W_{j,1} = A W_{j,0}$, and relative wages stay unchanged, $\frac{W_{i,0}}{W_{j,0}} = \frac{W_{i,1}}{W_{j,1}}$. Thus, wage-bill shares s_{it} and desired markdowns μ_{it} also remain the same: $s_{i,1} = s_{i,0}$ and $\mu_{i,1} = \mu_{i,0}$. Substituting these into equation (4.1) yields

$$W_{i,0}^* = \frac{z_i M_0}{\mu_{i,0}}, \quad A W_{i,0}^* = \frac{z_i M_1}{\mu_{i,0}}, \quad (\text{A.66})$$

which validates the conjecture for $A = M_1/M_0$. In sum, flexible wages imply complete and uniform passthrough across firms. ■

Auxiliary Lemmas for the Proof of Proposition 2.

LEMMA 2. *Suppose that $M_1 > M_0$ and firm m cannot adjust its wage, so that $W_{m1} = W_{m0}$. Then both adjusting firms increase their wage-bill shares:*

$$s_{i1} > s_{i0} \quad \text{for all } i \neq m.$$

Proof. Let l and s denote the two adjusting firms. Define

$$g(s) \equiv \frac{1}{\mu(s)} = \frac{\eta - (\eta - \zeta)s}{1 + \eta - (\eta - \zeta)s},$$

the inverse of the desired markdown, and let

$$X_t \equiv \frac{M_t}{W_{jt}}.$$

Using the static wage-setting equation and $s_i = (W_i/W_j)^{1+\eta}$, the optimality condition for an adjusting firm can be written as

$$s_{it}^{\frac{1}{1+\eta}} = g(s_{it})z_iX_t.$$

Equivalently,

$$H(s_{it}) = z_iX_t, \quad H(s) \equiv \frac{s^{\frac{1}{1+\eta}}}{g(s)}.$$

The function H is strictly increasing because $s^{1/(1+\eta)}$ is strictly increasing and $g(s)$ is strictly decreasing.

We first show that $X_1 > X_0$. Suppose instead that $X_1 \leq X_0$. Since H is strictly increasing, both adjusting firms must weakly decrease their wage-bill shares:

$$s_{l1} \leq s_{l0}, \quad s_{s1} \leq s_{s0}.$$

Because wage-bill shares sum to one, this implies

$$s_{m1} \geq s_{m0}.$$

But firm m does not adjust its wage, so $W_{m1} = W_{m0}$. Since

$$s_{mt} = \left(\frac{W_{mt}}{W_{jt}} \right)^{1+\eta},$$

the inequality $s_{m1} \geq s_{m0}$ implies $W_{j1} \leq W_{j0}$. Together with $M_1 > M_0$, this gives

$$X_1 = \frac{M_1}{W_{j1}} > \frac{M_0}{W_{j0}} = X_0,$$

a contradiction. Hence $X_1 > X_0$.

Since $H(s_{it}) = z_i X_t$ for each adjusting firm and H is strictly increasing, $X_1 > X_0$ implies

$$s_{i1} > s_{i0} \quad \text{for all } i \neq m.$$

■

LEMMA 3. *Consider two adjusting firms in sector j with $z_l > z_s$. Then $s_l > s_s$, and hence $W_l > W_s$ and $n_l > n_s$.*

Proof. Dividing the labor supply curves yields

$$\frac{n_l}{n_s} = \left(\frac{W_l}{W_s} \right)^\eta.$$

So we need to show that $W_l > W_s$, which is equivalent to $s_l > s_s$. Rearranging the optimal wage setting equation (4.1), we obtain

$$s_i^{\frac{1}{1+\eta}} = \frac{\eta - (\eta - \zeta)s_i}{1 + \eta - (\eta - \zeta)s_i} z_i \frac{M}{W_j}.$$

So that

$$\left(\frac{s_l}{s_s} \right)^{\frac{1}{1+\eta}} = \frac{1 - \frac{1}{1+\eta-(\eta-\zeta)s_l} z_l}{1 - \frac{1}{1+\eta-(\eta-\zeta)s_s} z_s} z_s. \quad (\text{A.67})$$

Suppose by contradiction then $s_s \geq s_l$. Then, we have that

$$\frac{1}{1 + \eta - (\eta - \zeta)s_l} \leq \frac{1}{1 + \eta - (\eta - \zeta)s_s},$$

so that

$$\frac{1 - \frac{1}{1+\eta-(\eta-\zeta)s_l} z_l}{1 - \frac{1}{1+\eta-(\eta-\zeta)s_s} z_s} \geq 1.$$

However, because $\frac{z_l}{z_s} > 1$, equation (A.67) implies that $s_l > s_s$ because the RHS of the equation is strictly greater than one. A contradiction. Thus, $s_l > s_s$. Since $s_i = (W_i/W_j)^{1+\eta}$, this implies $W_l > W_s$; and from $n_l/n_s = (W_l/W_s)^\eta$, it follows that $n_l > n_s$. ■

Proof of Proposition 2. Proof. Let l and s denote the two adjusting firms, with $z_l > z_s$. By Lemma 3, the more productive firm has the larger initial wage-bill share, so $s_{l0} > s_{s0}$.

For any variable X , let $\Delta X \equiv X_1/X_0$ denote the gross change from period 0 to period 1. Since

$$s_i = \left(\frac{W_i}{W_j} \right)^{1+\eta},$$

we have

$$\Delta \ln s_l - \Delta \ln s_s = (1 + \eta)(\Delta \ln W_l - \Delta \ln W_s).$$

Thus it is enough to show that $\Delta \ln W_l < \Delta \ln W_s$.

Suppose, toward a contradiction, that $\Delta \ln W_l \geq \Delta \ln W_s$. Then $\Delta s_l \geq \Delta s_s$. Define

$$g(s) \equiv \frac{1}{\mu(s)} = \frac{\eta - (\eta - \zeta)s}{1 + \eta - (\eta - \zeta)s},$$

the inverse of the desired markdown. Using the static wage-setting equation,

$$W_i = g(s_i)z_i M,$$

and dividing by the local wage index, we can write

$$s_i^{\frac{1}{1+\eta}} = g(s_i)z_i \frac{M}{W_j}.$$

Taking gross changes between periods 0 and 1 gives

$$\Delta s_i^{\frac{1}{1+\eta}} = \Delta g(s_i) \Delta \left(\frac{M}{W_j} \right).$$

Because $\Delta(M/W_j)$ is common across the two adjusting firms, the assumption $\Delta s_l \geq \Delta s_s$ implies

$$\Delta g(s_l) \geq \Delta g(s_s).$$

We now show that the opposite holds. Since

$$\Delta g(s_i) = \frac{\eta - (\eta - \zeta)s_{i1}}{\eta - (\eta - \zeta)s_{i0}} \frac{1 + \eta - (\eta - \zeta)s_{i0}}{1 + \eta - (\eta - \zeta)s_{i1}},$$

the inequality $\Delta g(s_l) < \Delta g(s_s)$ is equivalent to

$$\frac{s_{l1} - s_{l0}}{s_{s1} - s_{s0}} \frac{\eta - (\eta - \zeta)s_{s0}}{\eta - (\eta - \zeta)s_{l0}} \frac{1 + \eta - (\eta - \zeta)s_{s1}}{1 + \eta - (\eta - \zeta)s_{l1}} > 1.$$

By Lemma 2, $s_{s1} > s_{s0}$, so the denominator is strictly positive.

The maintained contradiction hypothesis implies

$$\frac{s_{l1}}{s_{l0}} \geq \frac{s_{s1}}{s_{s0}},$$

and since $s_{s1} > s_{s0}$, this gives

$$\frac{s_{l1} - s_{l0}}{s_{s1} - s_{s0}} \geq \frac{s_{l0}}{s_{s0}}.$$

It is therefore sufficient to show that

$$\frac{s_{l0}}{s_{s0}} \frac{\eta - (\eta - \zeta)s_{s0}}{\eta - (\eta - \zeta)s_{l0}} \frac{1 + \eta - (\eta - \zeta)s_{s1}}{1 + \eta - (\eta - \zeta)s_{l1}} > 1.$$

Equivalently,

$$\frac{\eta/s_{s0} - (\eta - \zeta)}{\eta/s_{l0} - (\eta - \zeta)} \times \frac{1 + \eta - (\eta - \zeta)s_{s1}}{1 + \eta - (\eta - \zeta)s_{l1}} > 1.$$

The first fraction is strictly greater than one because $s_{l0} > s_{s0}$. The second fraction is strictly greater than one because the maintained contradiction hypothesis and $s_{l0} > s_{s0}$ imply $s_{l1} > s_{s1}$. Hence

$$\Delta g(s_l) < \Delta g(s_s),$$

contradicting the implication above that $\Delta g(s_l) \geq \Delta g(s_s)$.

Therefore

$$\Delta \ln W_l < \Delta \ln W_s.$$

Since

$$\Delta \ln s_l - \Delta \ln s_s = (1 + \eta)(\Delta \ln W_l - \Delta \ln W_s),$$

we obtain

$$\Delta \ln s_l < \Delta \ln s_s.$$

Thus, among the adjusting firms, the smaller firm expands its wage-bill share by more. ■

B Solution Method

This appendix describes how to compute the Jacobians of the producer block. The Jacobians are conditional on the realization of Calvo shocks, which break the time symmetries that facilitate the fake news algorithm in Auclert et al. (2021). Nevertheless, the Jacobians can be computed efficiently, given that the wage setting equation is a well-behaved rootfinding problem with analytical derivatives. In particular, the Jacobians are available in closed form via the implicit function theorem, so we never resort to numerical differentiation.

This appendix assumes that the deterministic steady state of the model is known. The steady state is straightforward to compute, for example by iterating the static best responses from an initial guess that all firms set the same markdown.

B.1 Best Response Functions

The first step is to implement a function that maps the sequences $\{m_t, \pi_t, w_{jt}\}_{t=0}^T$ into best responses $\{\tilde{w}_{ijt}\}_{t=0}^T$ for all firms $i = 1, \dots, M_j$. We perform this step separately for all markets $j = 1, \dots, J$, and compute the implied local wage indices $\{\tilde{w}_{ijt}\}_{t=0}^T$ as well. The tilde signifies that the best responses are generally different from the input wage indices. Note that we do not include $\{W_t, N_t, i_t\}_{t=0}^T$ among the inputs of the best response function, even though they appear in the first-order condition, because they have no first-order effect on $\tilde{w}_{ijt}$.

Optimal reset wage. Let H_{ijt} denote the planning horizon of firm i on market j resetting its wage at time t , i.e. the length of wage spell $\mathcal{S}_{ijt}$. Given the assumption of perfect foresight, the wage setting equation (3.20) can be rewritten in real terms as

$$w_{ijt}^* = \frac{\sum_{\tau=0}^{H_{ijt}} R_{t,t+\tau} \pi_{t,t+\tau} n_{ijt+\tau} [\eta - (\eta - \zeta) s_{ijt+\tau}] m_{t+\tau} z_{ij}}{\sum_{\tau=0}^{H_{ijt}} R_{t,t+\tau} n_{ijt+\tau} [1 + \eta - (\eta - \zeta) s_{ijt+\tau}]} \quad (\text{B.1})$$

Substituting the labor supply curve and the expression for market share yields

$$w_{ijt}^* = \frac{\sum_{\tau=0}^{H_{ijt}} R_{t,t+\tau} \pi_{t,t+\tau} \left[\eta - (\eta - \zeta) \left(\frac{w_{ijt}^*}{w_{jt+\tau} \pi_{t,t+\tau}} \right)^{1+\eta} \right] m_{t+\tau} z_{ij} \left[\left(\frac{w_{ijt}^*}{w_{jt+\tau} \pi_{t,t+\tau}} \right)^\eta \left(\frac{w_{jt+\tau}}{w_{t+\tau}} \right)^\zeta N_{t+\tau} \right]}{\sum_{\tau=0}^{H_{ijt}} R_{t,t+\tau} \left[1 + \eta - (\eta - \zeta) \left(\frac{w_{ijt}^*}{w_{jt+\tau} \pi_{t,t+\tau}} \right)^{1+\eta} \right] \left(\frac{w_{ijt}^*}{w_{jt+\tau} \pi_{t,t+\tau}} \right)^\eta \left(\frac{w_{jt+\tau}}{w_{t+\tau}} \right)^\zeta N_{t+\tau}} \quad (\text{B.2})$$

This is a nonlinear equation for w_{ijt}^* , whose derivatives we have to characterize. Collecting the coefficients on the unknown, we can write

$$w_{ijt}^* = \frac{z_{ij} \cdot (w_{ijt}^*)^\eta \cdot A_{ijt} - z_{ij} \cdot (w_{ijt}^*)^{1+2\eta} \cdot B_{ijt}}{(w_{ijt}^*)^\eta \cdot C_{ijt} - (w_{ijt}^*)^{1+2\eta} \cdot D_{ijt}}, \quad (\text{B.3})$$

where the coefficients are

$$A_{ijt} = \sum_{\tau=0}^{H_{ijt}} R_{t+\tau} \pi_{t,t+\tau}^{1-\eta} \eta m_{t+\tau} \left[\left(\frac{1}{w_{jt+\tau}} \right)^\eta \left(\frac{w_{jt+\tau}}{W_{t+\tau}} \right)^\zeta N_{t+\tau} \right], \quad (\text{B.4})$$

$$B_{ijt} = \sum_{\tau=0}^{H_{ijt}} R_{t+\tau} \pi_{t,t+\tau}^{-2\eta} (\eta - \zeta) m_{t+\tau} \left(\frac{1}{w_{jt+\tau}} \right)^{1+\eta} \left[\left(\frac{1}{w_{jt+\tau}} \right)^\eta \left(\frac{w_{jt+\tau}}{W_{t+\tau}} \right)^\zeta N_{t+\tau} \right], \quad (\text{B.5})$$

$$C_{ijt} = \sum_{\tau=0}^{H_{ijt}} R_{t+\tau} \pi_{t,t+\tau}^{-\eta} (1 + \eta) \left(\frac{1}{w_{jt+\tau}} \right)^\eta \left(\frac{w_{jt+\tau}}{W_{t+\tau}} \right)^\zeta N_{t+\tau}, \quad (\text{B.6})$$

$$D_{ijt} = \sum_{\tau=0}^{H_{ijt}} R_{t+\tau} \pi_{t,t+\tau}^{-(1+2\eta)} (\eta - \zeta) \left(\frac{1}{w_{jt+\tau}} \right)^{1+2\eta} \left(\frac{w_{jt+\tau}}{W_{t+\tau}} \right)^\zeta N_{t+\tau}. \quad (\text{B.7})$$

Let's express the wage setting equation and its derivative as

$$F(w_{ijt}^*) = z_{ij} \left[(w_{ijt}^*)^\eta A_{ijt} - (w_{ijt}^*)^{1+2\eta} B_{ijt} \right] - \left[(w_{ijt}^*)^{1+\eta} C_{ijt} - (w_{ijt}^*)^{2+2\eta} D_{ijt} \right], \quad (\text{B.8})$$

$$F'(w_{ijt}^*) = z_{ij} \left[\eta (w_{ijt}^*)^{\eta-1} A_{ijt} - [1 + 2\eta] (w_{ijt}^*)^{2\eta} B_{ijt} \right] - \left[(1 + \eta) (w_{ijt}^*)^\eta C_{ijt} - (2 + 2\eta) (w_{ijt}^*)^{1+2\eta} D_{ijt} \right], \quad (\text{B.9})$$

which we use to compute the derivatives of w_{ijt}^* . By the implicit function theorem, the response of the reset wage to any input $x \in \{w_{j0}, \dots, w_{jT}, \pi_0, \dots, \pi_T, m_0, \dots, m_T\}$ is

$$\frac{\partial w_{ijt}^*}{\partial x} = -\frac{1}{F'(w_{ijt}^*)} \left[z_{ij} (w_{ijt}^*)^\eta \frac{\partial A_{ijt}}{\partial x} - z_{ij} (w_{ijt}^*)^{1+2\eta} \frac{\partial B_{ijt}}{\partial x} - (w_{ijt}^*)^{1+\eta} \frac{\partial C_{ijt}}{\partial x} + (w_{ijt}^*)^{2+2\eta} \frac{\partial D_{ijt}}{\partial x} \right]. \quad (\text{B.10})$$

Realized wage. The actual real wage of firm i on market j is

$$\tilde{w}_{ijt} = \begin{cases} w_{ijt}^* & \text{if } \Theta_{ijt} = 1, \\ \frac{\tilde{w}_{ijt-1}}{\pi_t} & \text{if } \Theta_{ijt} = 0. \end{cases} \quad (\text{B.11})$$

Therefore, its perturbations can be obtained iteratively:

$$\frac{\partial \tilde{w}_{ijt}}{\partial x} = \begin{cases} \frac{\partial w_{ijt}^*}{\partial x} & \Theta_{ijt} = 1, \\ \frac{1}{\pi_{ss}} \frac{\partial \tilde{w}_{ijt-1}}{\partial x} - \frac{\tilde{w}_{ij,ss}}{\pi_{ss}^2} \frac{\partial \pi_t}{\partial x} & \Theta_{ijt} = 0. \end{cases} \quad (\text{B.12})$$

Aggregation. Given sequences of firm level wages $\{\tilde{w}_{ijt}\}$, the local wage index is given by

$$\tilde{w}_{jt} = \left(\sum_{i=1}^{M_j} \tilde{w}_{ijt}^{1+\eta} \right)^{\frac{1}{1+\eta}}. \quad (\text{B.13})$$

Differentiating it at the steady state gives

$$\frac{\partial \tilde{w}_{jt}}{\partial x} = \tilde{w}_{j,ss}^{-\eta} \sum_{i=1}^{M_j} \tilde{w}_{ij,ss}^{\eta} \frac{\partial \tilde{w}_{ijt}}{\partial x}, \quad (\text{B.14})$$

To summarize, we have constructed the partial derivatives of the best-response functions for each market in sequence space. Letting bold letters denote sequences, we may write the best-response function as $\tilde{\mathbf{w}}_j = \mathbf{F}_j(\mathbf{w}_j, \boldsymbol{\pi}, \mathbf{m})$.

B.2 Nash Equilibrium on Local Markets

Nash equilibrium is the fixed point $\tilde{\mathbf{w}}_j = \mathbf{w}_j$. By the implicit function theorem, the Jacobian of the Nash-equilibrium wage with respect to either input $\mathbf{x} \in \{\boldsymbol{\pi}, \mathbf{m}\}$, evaluated at the steady state, is

$$\frac{d\mathbf{w}_j}{d\mathbf{x}} = - \left(\frac{\partial \mathbf{F}_j}{\partial \mathbf{w}_j} - \mathbf{I} \right)^{-1} \frac{\partial \mathbf{F}_j}{\partial \mathbf{x}}. \quad (\text{B.15})$$

We compute these “solved Jacobians” separately for each market, and then obtain the Jacobian of the aggregate wage index as

$$\frac{d\mathbf{w}}{d\mathbf{x}} = w_{ss}^{-\zeta} \left(\frac{1}{J} \sum_{j=1}^J w_{j,ss}^{\zeta} \frac{d\mathbf{w}_j}{d\mathbf{x}} \right). \quad (\text{B.16})$$

B.3 Jacobians of TFP

Let us define TFP A_t such that $Y_t = A_t N_t$. Our goal is to compute the Jacobians of $\{A_t\}_{t=0}^T$ with respect to $\{\pi_t, m_t\}_{t=0}^T$ under Nash equilibrium on local markets. Aggregate output is

$$Y_t = \frac{1}{J} \sum_{j=1}^J \sum_{i=1}^{M_j} z_{ij} n_{ijt} = \frac{1}{J} \sum_{j=1}^J \sum_{i=1}^{M_j} z_{ij} \left(\frac{w_{ijt}}{w_{jt}} \right)^{\eta} \left(\frac{w_{jt}}{w_t} \right)^{\zeta} N_t. \quad (\text{B.17})$$

Therefore TFP is

$$A_t = \frac{Y_t}{N_t} = \frac{1}{J} \sum_{j=1}^J \sum_{i=1}^{M_j} z_{ij} \left(\frac{w_{ijt}}{w_{jt}} \right)^{\eta} \left(\frac{w_{jt}}{w_t} \right)^{\zeta}. \quad (\text{B.18})$$

As in the case of wage strategies, we want to compute a sufficient statistic A_{jt} separately for each market. The natural candidate is

$$A_{jt} = \sum_{i=1}^{M_j} z_{ij} \left(\frac{w_{ijt}}{w_{jt}} \right)^{\eta} \left(\frac{w_{jt}}{w_t} \right)^{\zeta} \quad (\text{B.19})$$

Conceptually, this is a function in sequence space $\mathbf{A}_j(\mathbf{w}_j, \mathbf{w}, \mathbf{m}, \boldsymbol{\pi})$, and we can use the Jacobians of Nash equilibrium wages to compute

$$\frac{d\mathbf{A}_j}{d\mathbf{x}} = \frac{\partial \mathbf{A}_j}{\partial \mathbf{x}} + \frac{\partial \mathbf{A}_j}{\partial \mathbf{w}_j} \frac{d\mathbf{w}_j}{d\mathbf{x}} + \sum_{j=1}^J \frac{\partial \mathbf{A}_j}{\partial \mathbf{w}} \frac{d\mathbf{w}}{d\mathbf{x}} \quad (\text{B.20})$$

It is simpler and more efficient however, to work with $B_{jt} = A_{jt}w_t^\zeta$ whose Jacobian can be computed market by market as

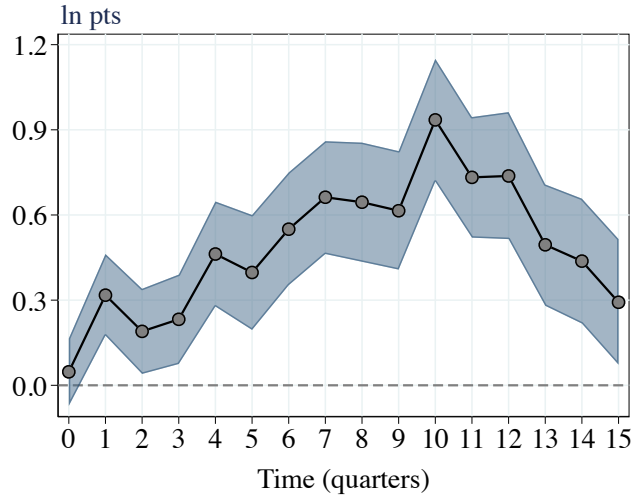
$$\frac{d\mathbf{B}_j}{d\mathbf{x}} = \frac{\partial\mathbf{B}_j}{\partial\mathbf{x}} + \frac{\partial\mathbf{B}_j}{\partial\mathbf{w}_j} \frac{d\mathbf{w}_j}{d\mathbf{x}} \quad (\text{B.21})$$

and then aggregated to obtain

$$\frac{d\mathbf{A}}{d\mathbf{x}} = \frac{1}{J} \sum_{j=1}^J \left(w_{ss}^{-\zeta} \cdot \frac{d\mathbf{B}_j}{d\mathbf{x}} - B_{j,ss} \cdot \zeta w_{ss}^{-\zeta-1} \frac{d\mathbf{w}}{d\mathbf{x}} \right). \quad (\text{B.22})$$

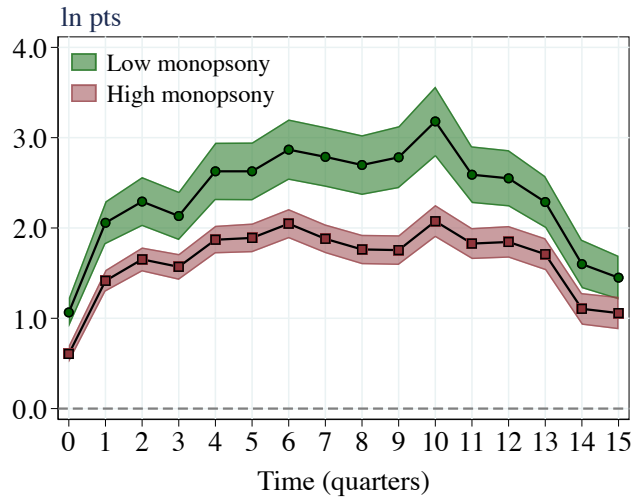
C Additional Figures and Tables

Figure C.1: WAGE BILL RESPONSE: MOVING AVERAGE VS. STANDARD LOCAL PROJECTION



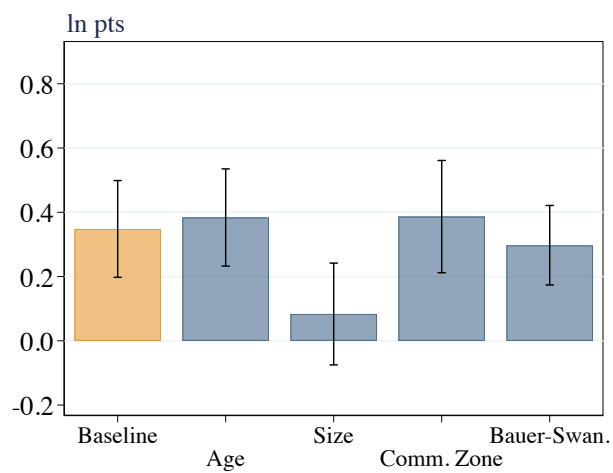
The figure compares the baseline moving average local projection with a standard local projection. The shaded areas represent 95 percent confidence intervals. Standard errors clustered at the local labor market level.

Figure C.2: WAGE BILL RESPONSE: HIGH VS. LOW MONOPSONY POWER

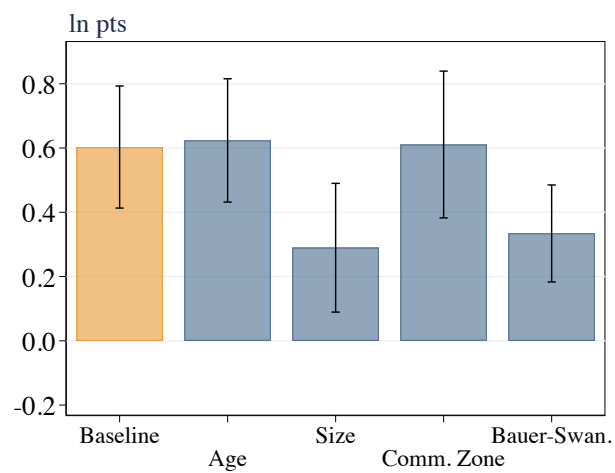


The figure shows the within-group response of the wage bill of high and low monopsony power firms to an expansionary monetary policy shock of 25 basis points. The shaded areas represent 95 percent confidence intervals. Standard errors clustered at the local labor market level.

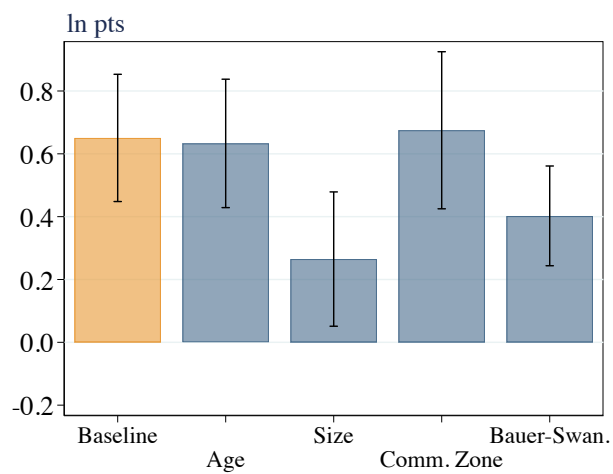
Figure C.3: ROBUSTNESS: DIFFERENT SPECIFICATION



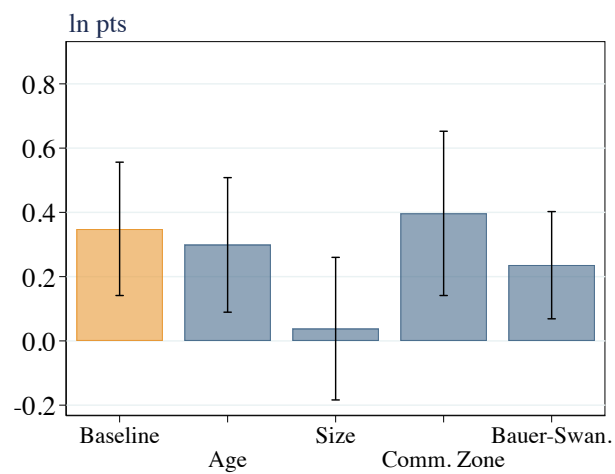
(a) Year 1



(b) Year 2

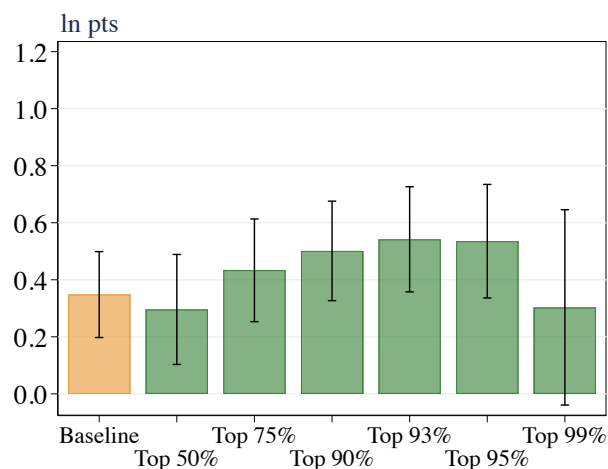


(c) Year 3

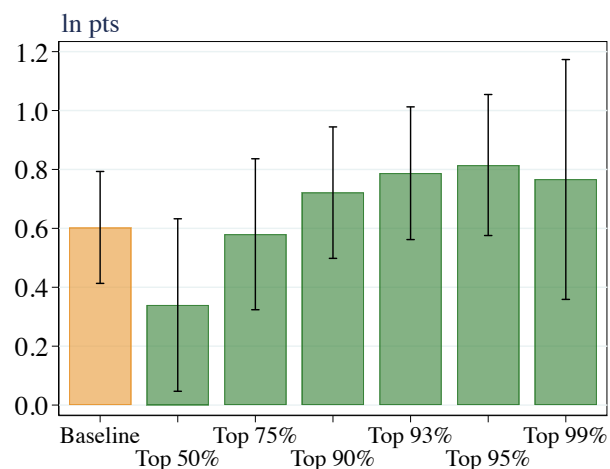


(d) Year 4

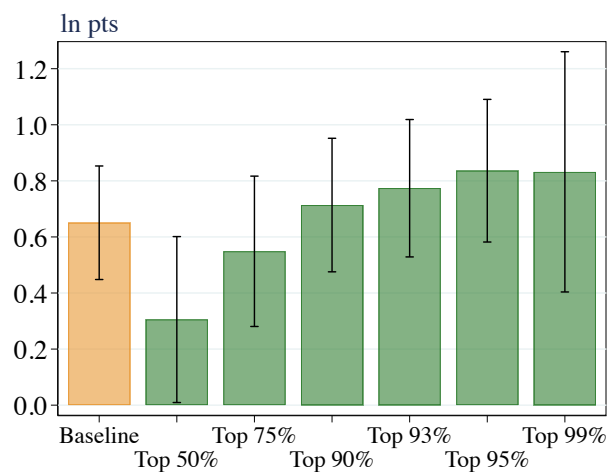
Figure C.4: ROBUSTNESS: DIFFERENT THRESHOLDS



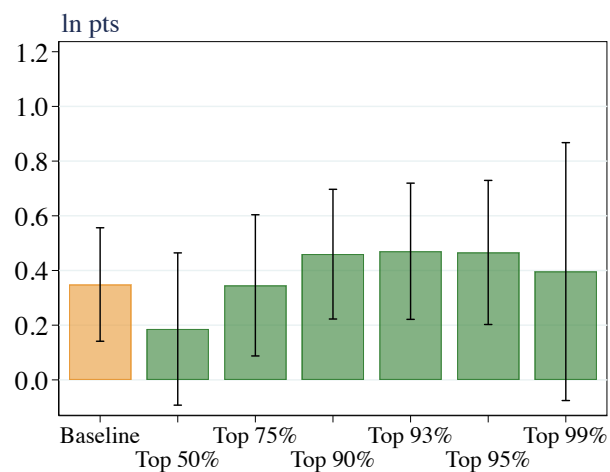
(a) Year 1



(b) Year 2

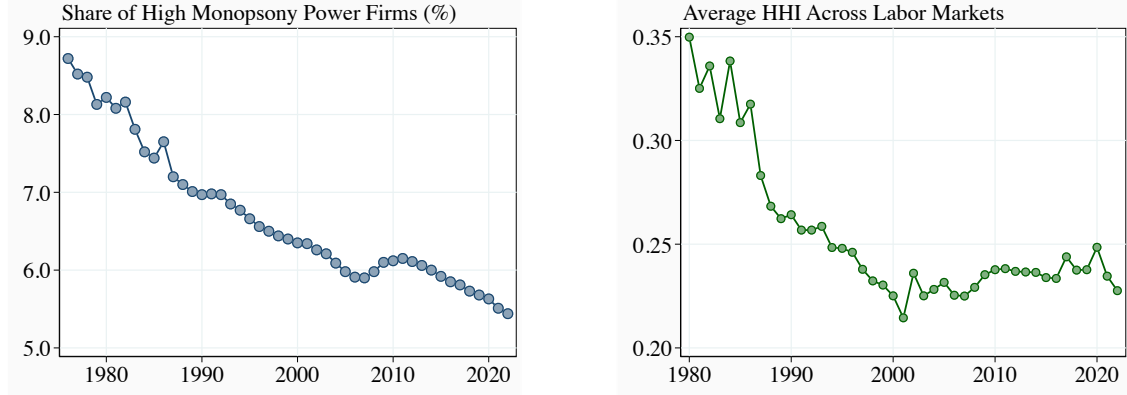


(c) Year 3



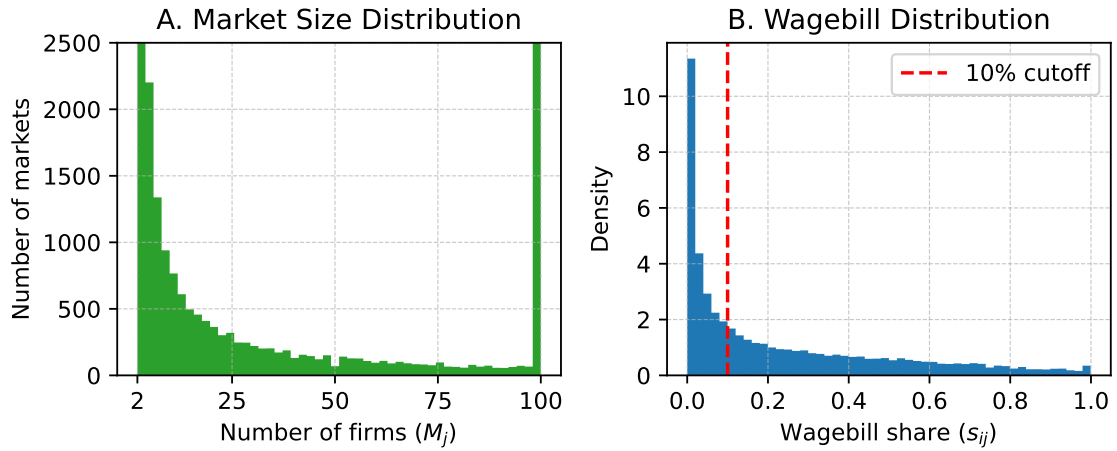
(d) Year 4

Figure C.5: LOCAL LABOR MARKET CONCENTRATION IN THE UNITED STATES



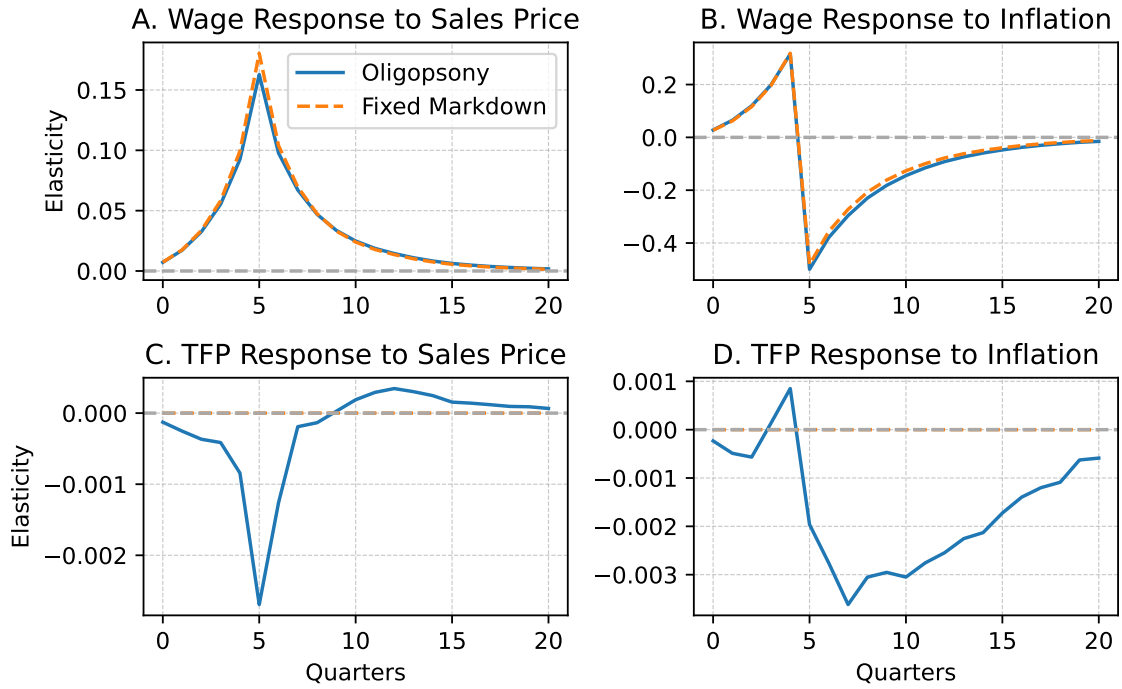
Note: Figure shows the evolution of local labor market concentration in the United States. We report two different measures, both of which are computed from the LBD. The left panel shows the share of high-monopsony firms, defined as having a market share of 10 percent or above. The right panel shows the average HHI across local labor markets.

Figure C.6: STEADY-STATE DISTRIBUTIONS



Note: Figure shows the distributions of the number of firms per market and the local wage bill shares of firms in the steady state of our model. The model has 10,000 local markets inhabited by 390,000 firms. Panel A shows that the number of firms per market range from 2 to a 100. Panel B shows that the wage bill share ranges from almost zero to 0.999, with the majority of firms having a wage bill share of less than 10 percent.

Figure C.7: FIRM BLOCK JACOBIANS



Note: Figure visualizes the differences between the Jacobians of the firm block $\{m_t, \pi_t\} \mapsto \{w_t, A_t\}$ under oligopsonistic competition and fixed desired markdowns. Each line may be interpreted as an impulse response to a one-time increase in m_t or π_t that is announced in period 0. For example, panel A shows $\frac{\partial \bar{w}_t}{\partial m_5} \frac{m_{ss}}{\bar{w}_{ss}}$ for $t = 0, 1, \dots, 20$.